

Future of BioMedical Implantable Antennas

Charles Langer, Daniel Gould, Grant Andrews, Nicholas
Allman, David Moreno

Current Implantable Antennas

- Current Antennas in MedRadio band (401-406MHz) are ~ 3-4 cm diameter
 - Constrained by battery size
- Examples:
 - Pacemakers, Brain Implants, Continuous Glucose Monitors, gastric stimulators, foot drop implants, cochlear implants
 - (check Chris Trampel, Lens into the body RF day 2017 in lab archives)

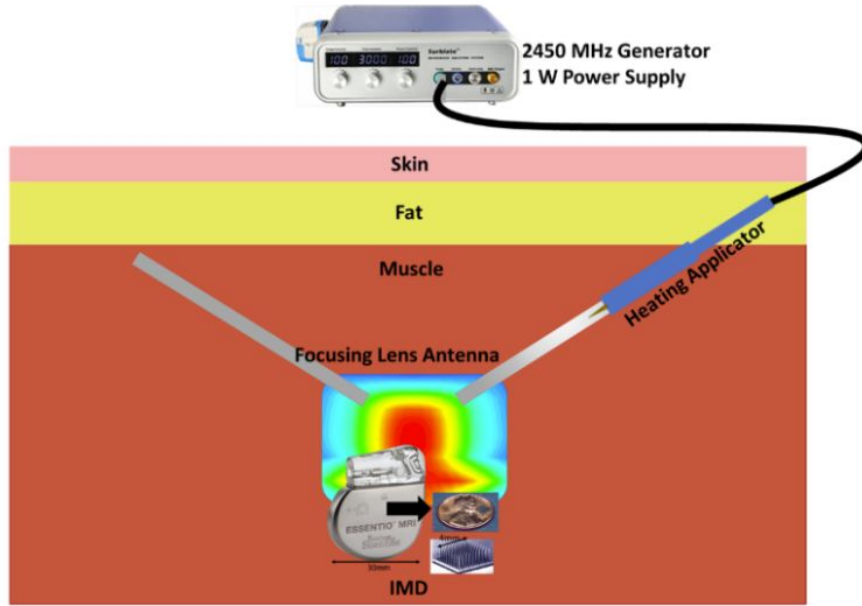
Future Implantable Antennas

- Battery technology improvement will allow for smaller medical implants
- Our antennas will be able to be reduced to ~1cm in diameter
 - These will be inefficient antennas
 - A possible fix will be adding focusing antennas to improve Rx/Tx signal strength

History of Passive Bio Compatible Focusing Antennas

- Implantable Wireless Antennas have been researched and used since the 1990s
- Researchers at the University of Utah started work on focusing implantable biocompatible antennas at the end of the 2010s
- An early research paper from the ECE School of Engineering titled “Field Focusing for Implanted Devices” was published in 2019
- The paper describes a method for electromagnetic field focusing used for 3D implantable devices by using two passive focusing antennas that guide the incident wave deep into the body to achieve a power increase of 8 to 16 times larger than methods where no focusing passive antennas are used

History of Passive Bio Compatible Focusing Antennas



Visual representation
of passive
biocompatible focusing
Antennas

Image taken from:

<https://utah.instructure.com/courses/558911/pages/bioelectromagnetics-implantable-antennas-and-more>

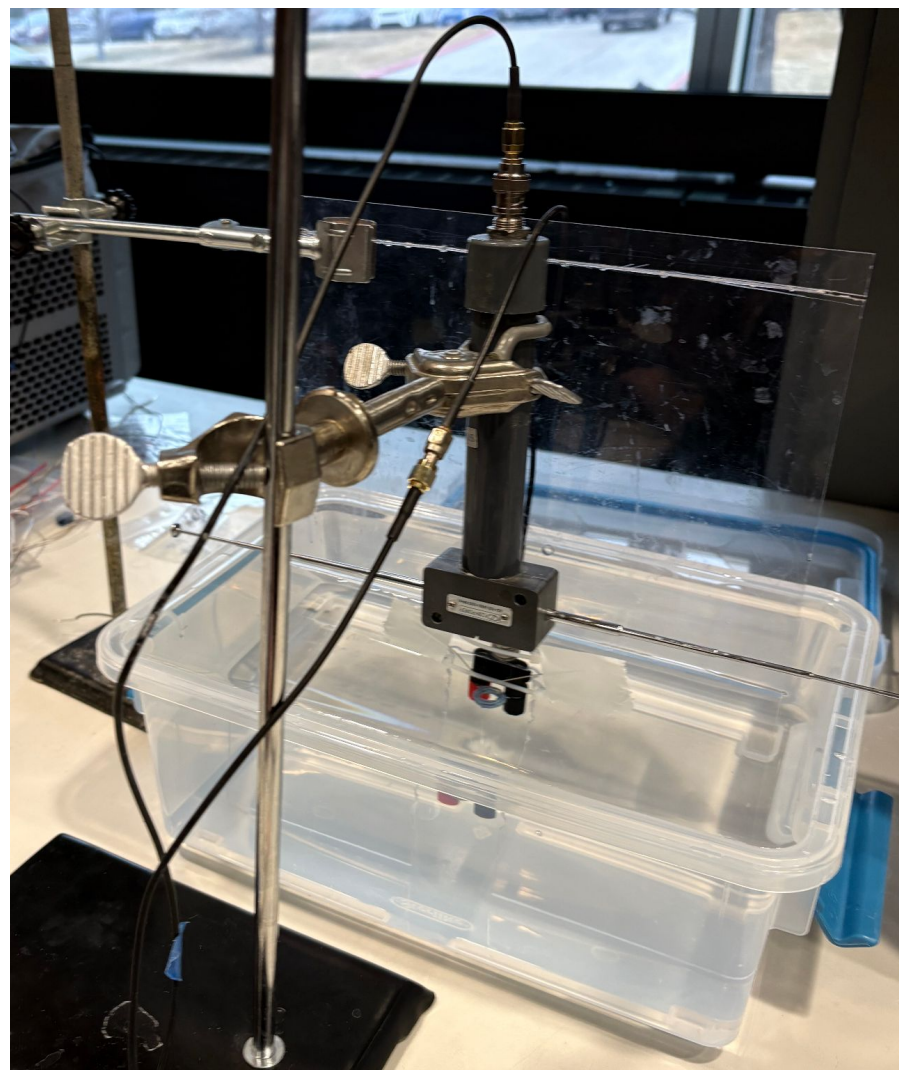
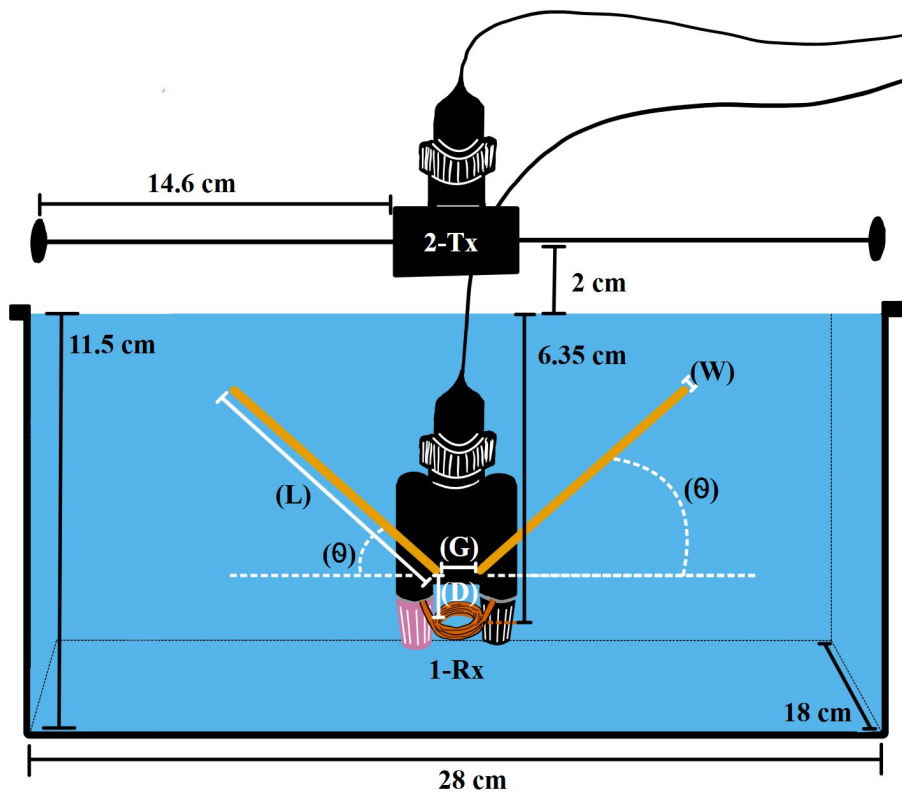
Our project goal

- Now we're trying to add a body compatible floppy wire through a cannula that will act as a passive wire to help with the Rx/Tx of a loop antenna in the muscle and transmitting antenna above the skin, fat, and muscle
- Why it should work?
 - These antennas rely on the propensity of electromagnetic fields to concentrate on points and gaps, and the ability of the antenna in the implant to pick up these stronger fields

Our Project

- Our set-up (Saline)
 - Put the picture of Raksha's set-up and edit it if we need
 - Sweep Range
 - Receiver dB without Passive antennas
 - Coil diameter, loops, how we made it (wrapped around expo marker)
 - Transmitter dB
 - Type of transmitter, length of L dipole, dist above water
 - Passive antennas
 - Length, gauge, angle, distance from center of Rx(S11) to lowest point?
 - Material of Antenna
 - copper
 - 1) liquid gallium encapsulated in flexible polyether block amide (PEBA) tubing
 - 2) injectable, metal-doped, conductive, thermoresponsive Poly Hydrogel (PNIPAM)
 - 3) polymer matrix composite of silver flakes and a biocompatible silicone adhesive, coated with Parylene C
 - S21 dB and why we care about it

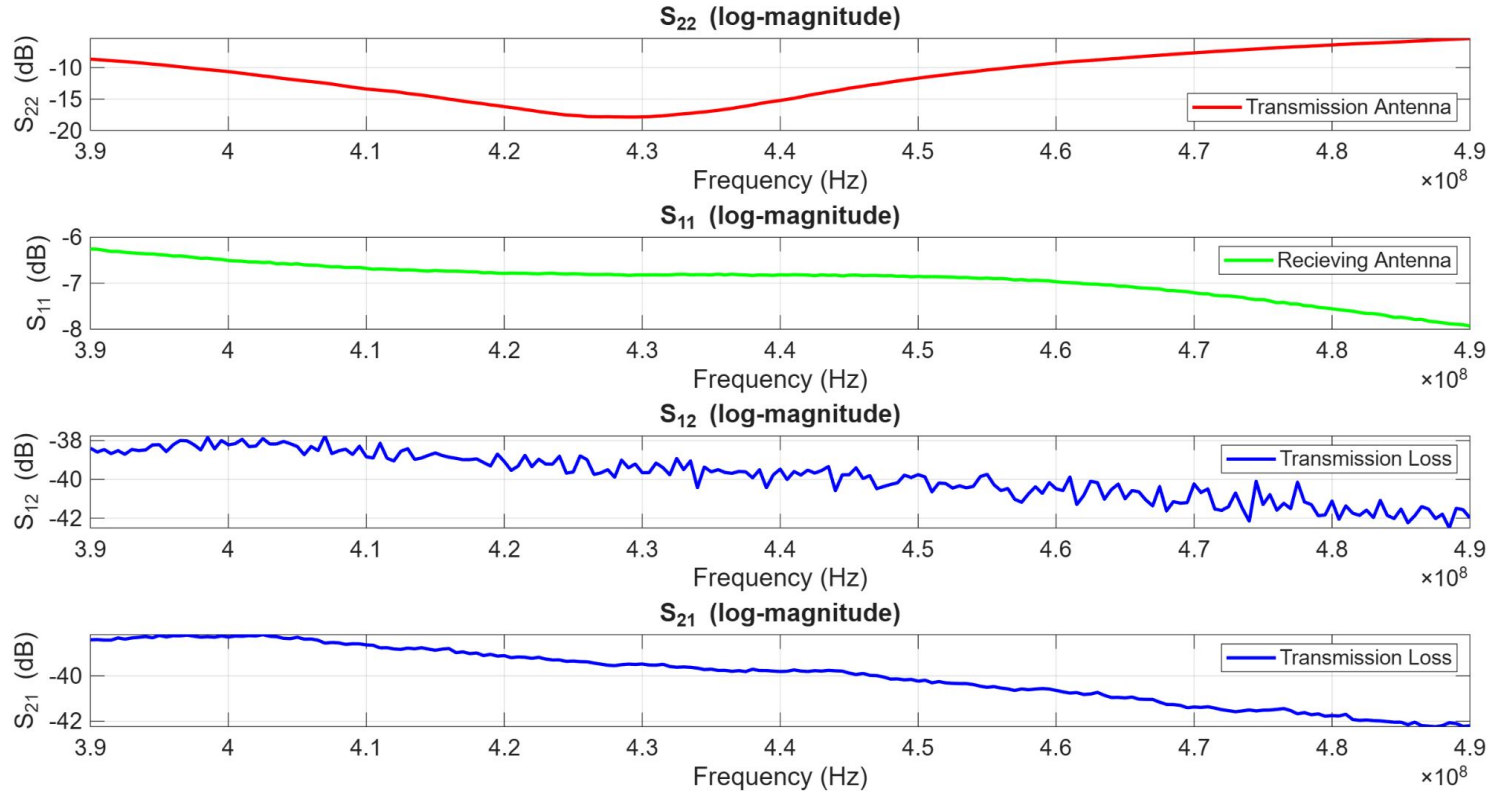
Our Test Setup



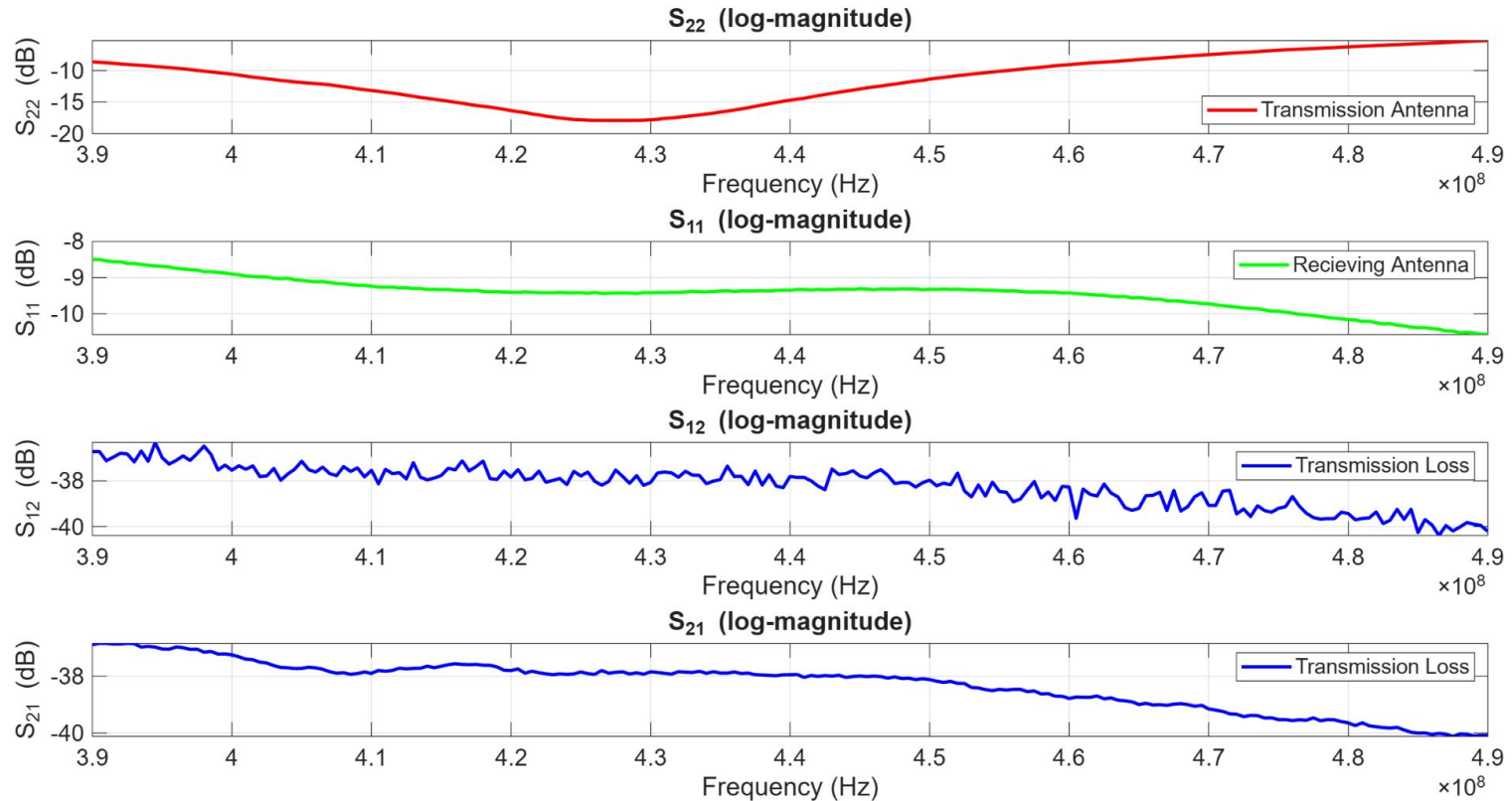
Measurements

- Recorded data from the VNA
 - dB of s_{11}
 - dB of s_{22}
 - dB of s_{21} or s_{12}
 - Primary one we are looking at so as to see if the passive antennas have an effect on transmission and reception
 - All measurements are taken between 390M Hz and 490M Hz, targeting 433M Hz.
 - Each test is set up in the same way every time, attempting to minimize outside influence and human error in our tests.

Representation of Full Data Range Taken - Reference Data



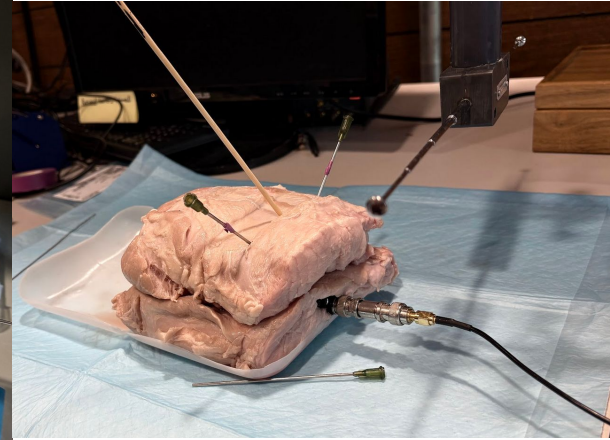
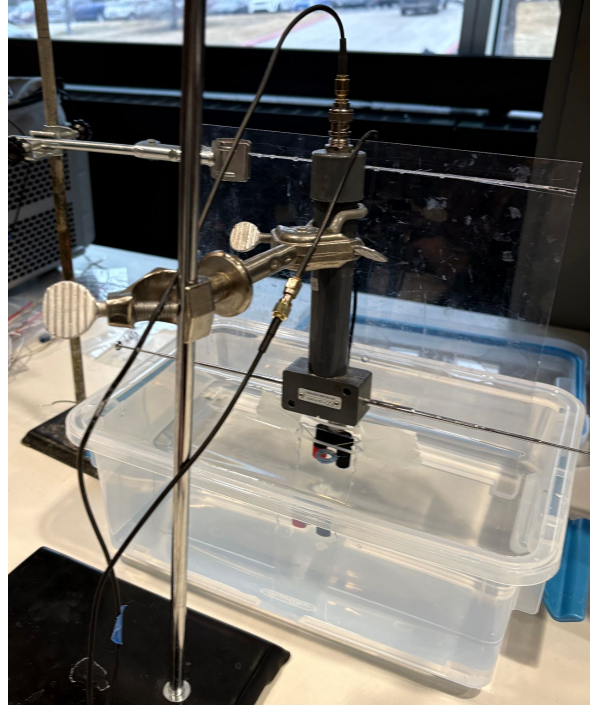
Representation of Full Data Range Taken - With Passive Ant.



Why Tested in Saline and Pork

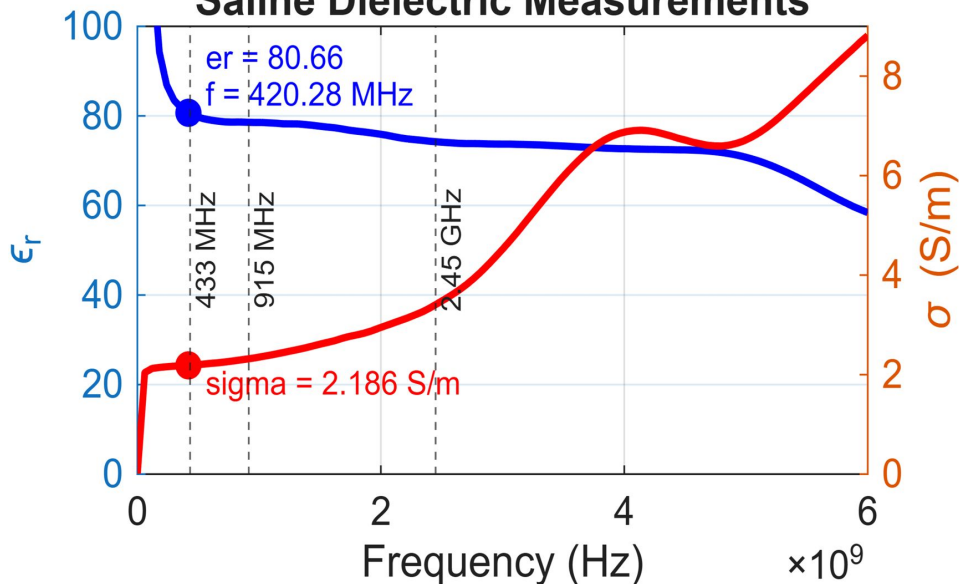
Saline has similar dielectric properties to blood and body tissues while allowing for manipulation of passive antennas

Pork has the most similar Dielectric properties to human tissue where these antennas are intended to be used.



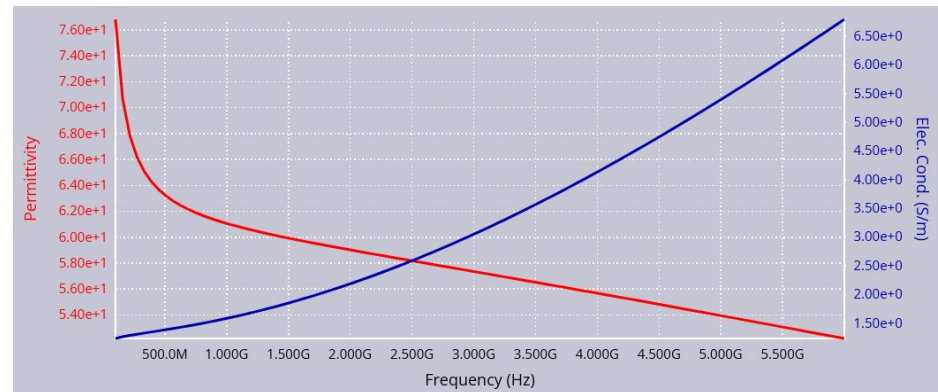
Dielectric Properties of Saline

Saline Dielectric Measurements

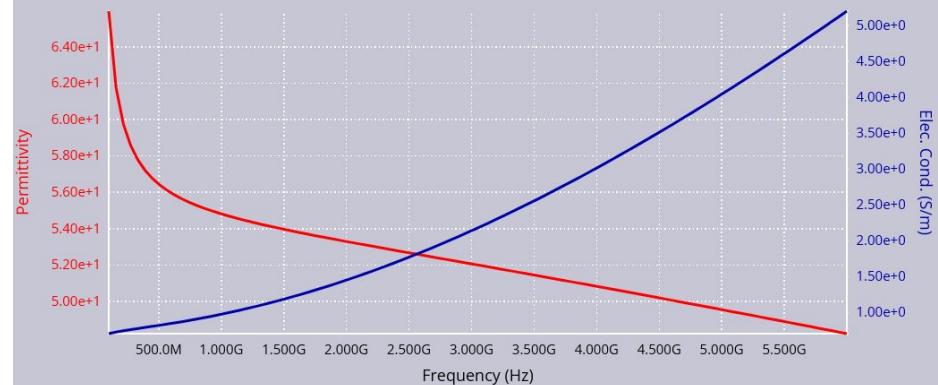


Fluctuations in Saline measurement are from inconsistencies in the measurement process or form a non-perfect calibration

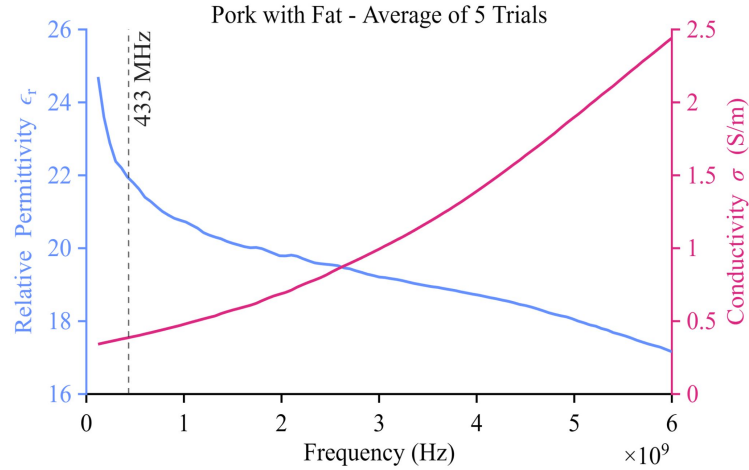
Dielectric Properties of Human Blood



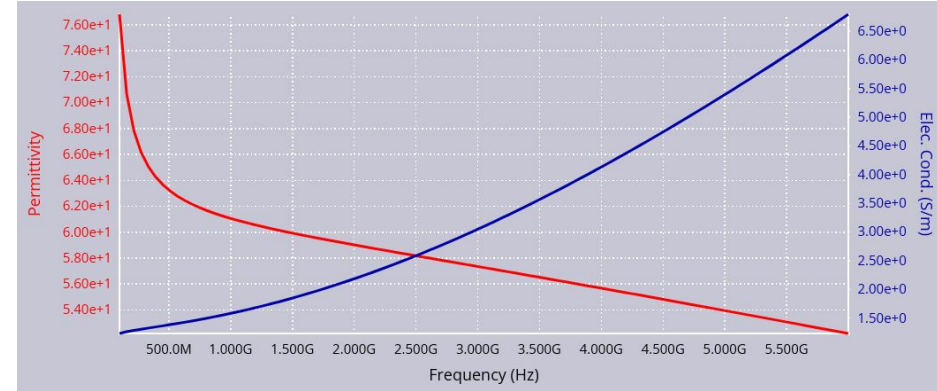
Dielectric Properties of Human Muscle



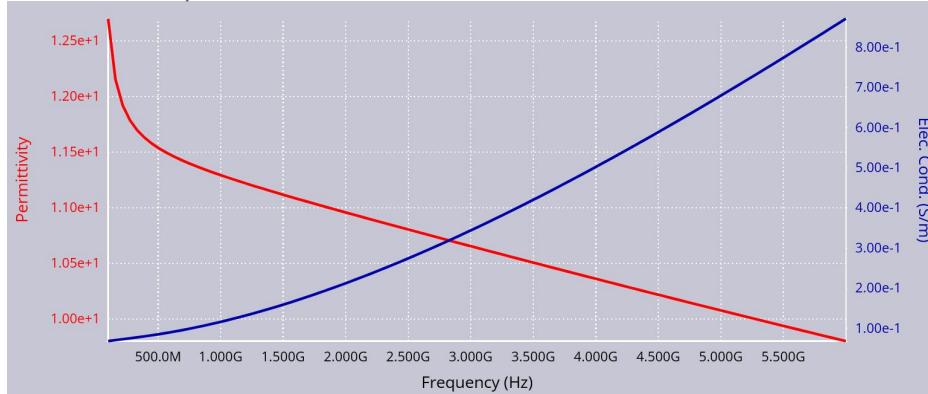
Dielectric Properties of Pork



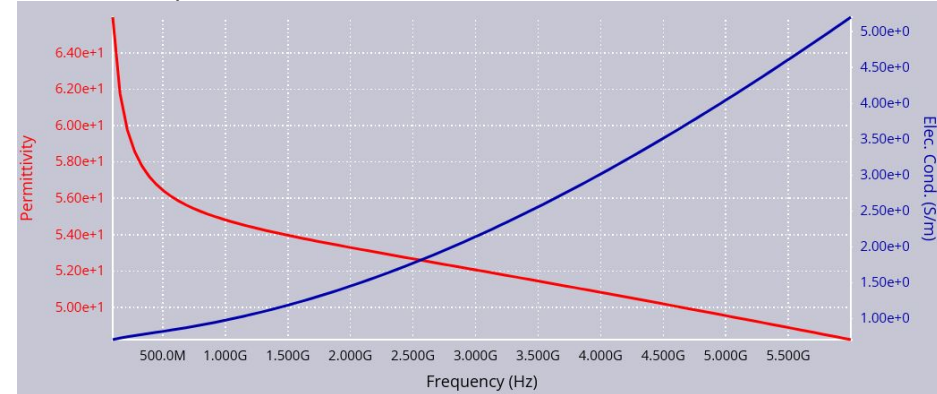
Dielectric Properties of Human Blood



Dielectric Properties of Human Fat



Dielectric Properties of Human Muscle

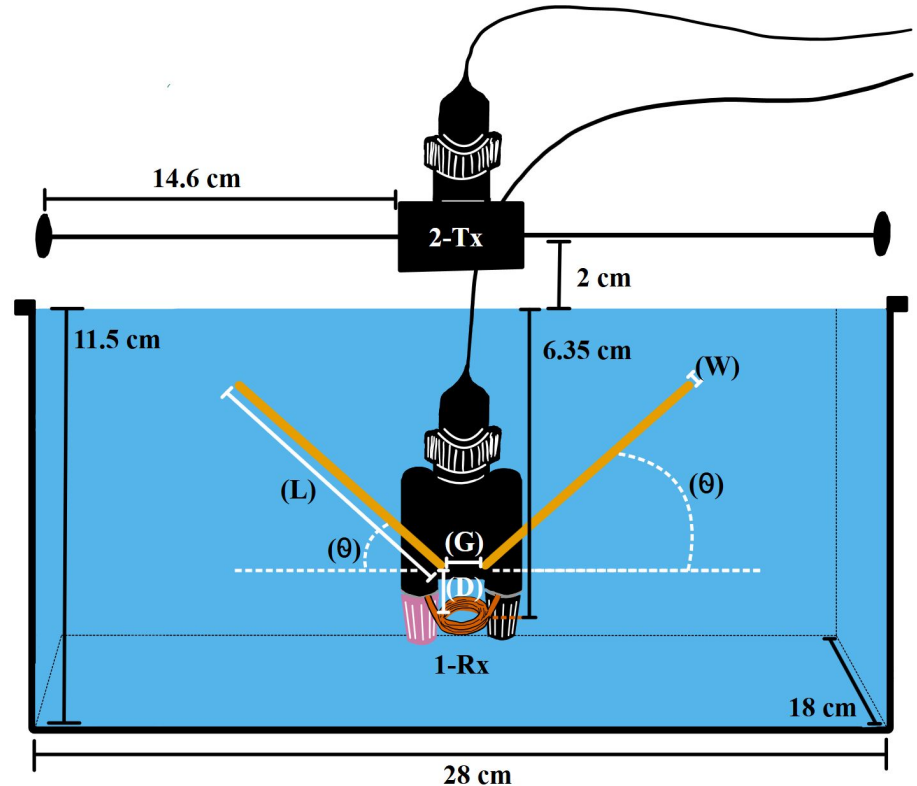


Physical Passive Antenna Parameters Tested in Saline

- Angle (θ) (Degrees)
- Gap (G) (cm)
- Distance to receiving Antenna (D) (cm)
- Length (L)(cm)
- Diameter (W)(mm)

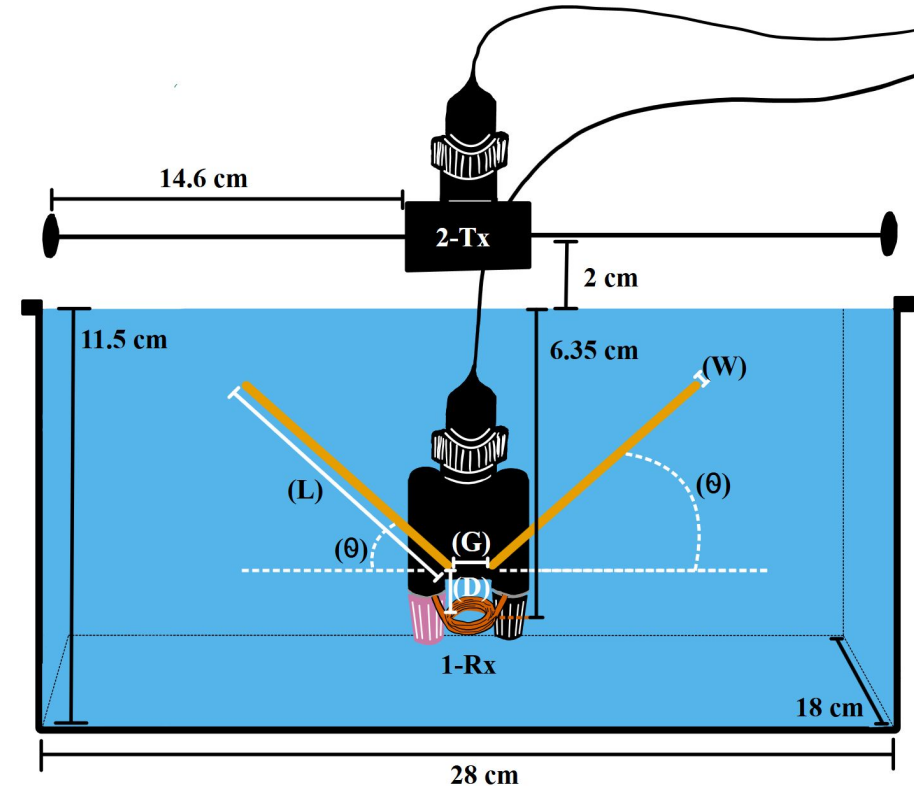
Parameter Effectiveness
interconnectivity:

- Length Vs. Angle



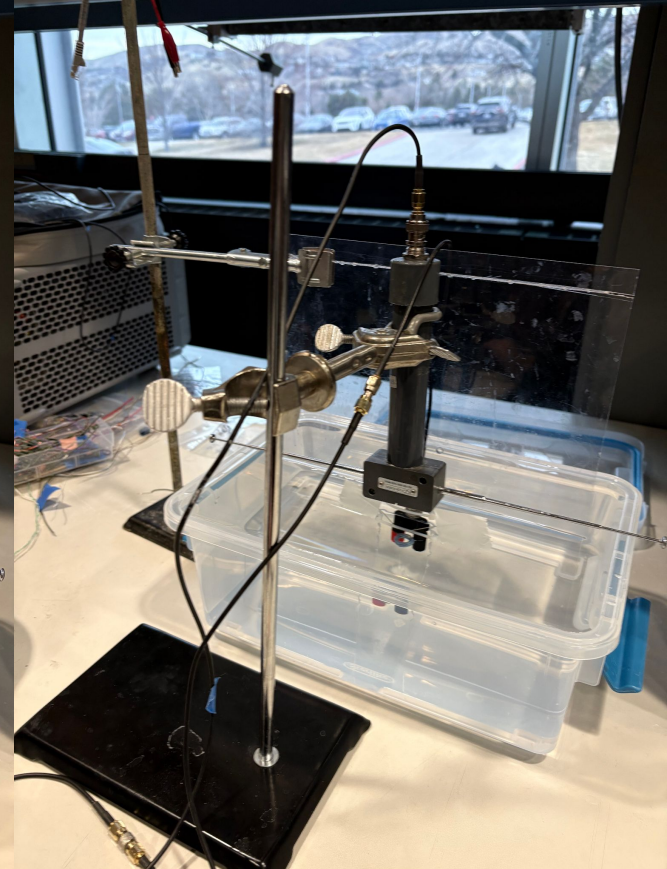
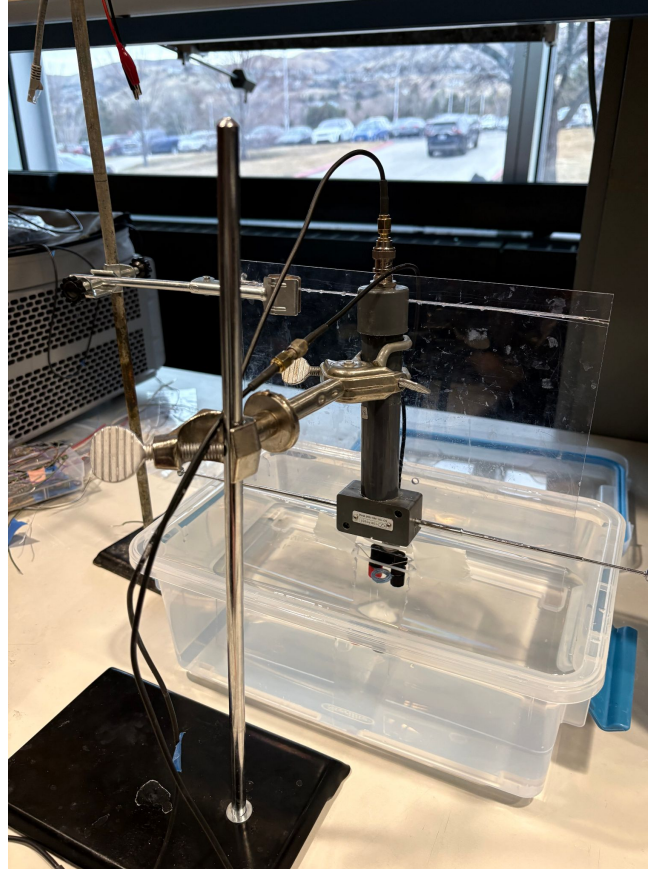
Copper Antenna Testing Method

- For our first set of Antenna Measurements we set the four variables not being tested to these set values:
 - $D = 1.0 \text{ cm}$
 - $G = 1.0 \text{ cm}$
 - $L = 2 \text{ in} = 5.08 \text{ cm}$
 - $\theta = 45 \text{ degrees}$
 - $W = 1.0 \text{ mm}$
- Each singular parameter was then varied to determine its best value relative to the other set values.
- A second series of (Optimal) tests were then conducted to confirm if the best value for each test variable stayed the same with all of the other parameters set to their preliminary optimal findings.
- One extra test determined the relationship between angle and length while keeping the length as long as possible while still in the saline for the angle tested.



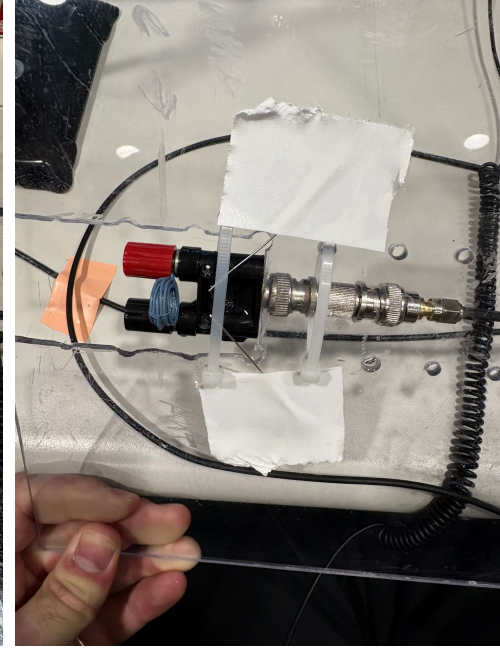
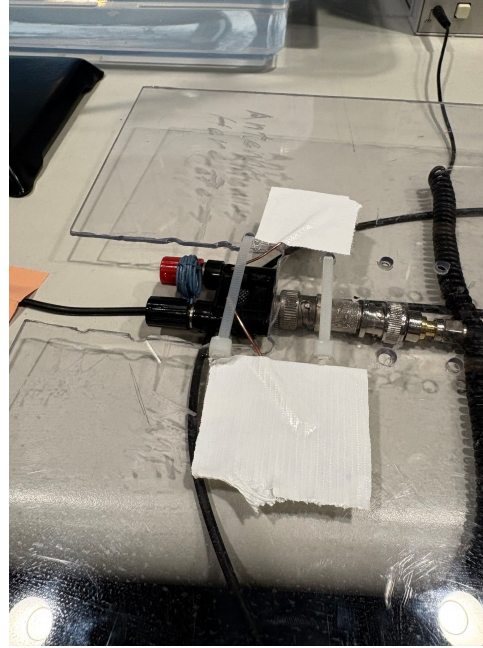
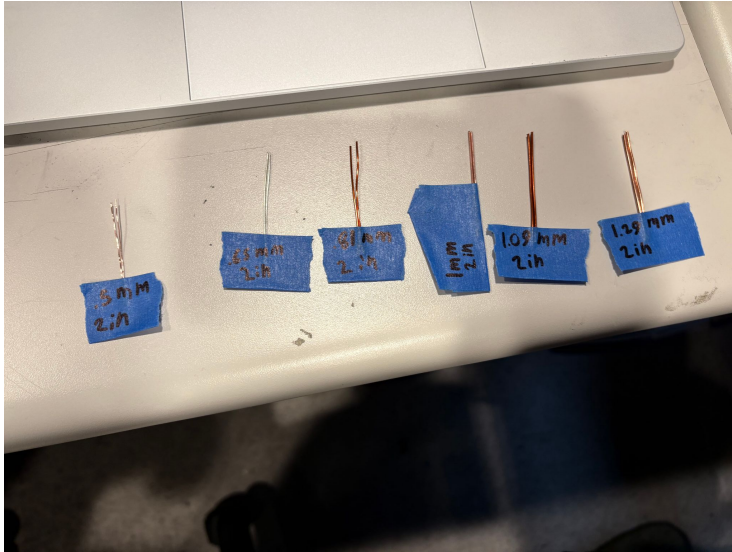
Test Variability

- We must remove the receiving antenna to place the passive antennas.
- This results in movement of the S parameter wires and can significantly change the transmission loss by up to 1 dB.
- As such we took multiple measurements with the S parameter antenna wires in many different locations
 - Trying to find the maximum and minimum values for each test.



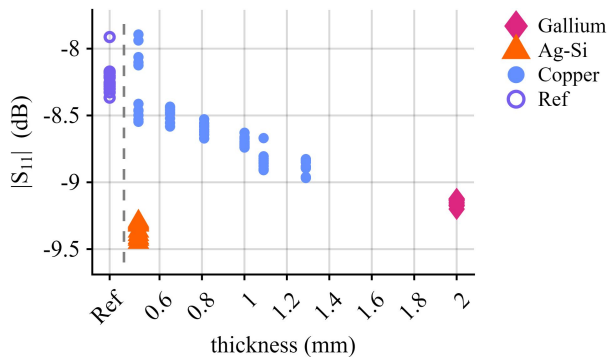
Copper Antenna Testing Method

Copper antennas were cut to length and then oriented on the plexiglass rig attached to the receiving antenna.

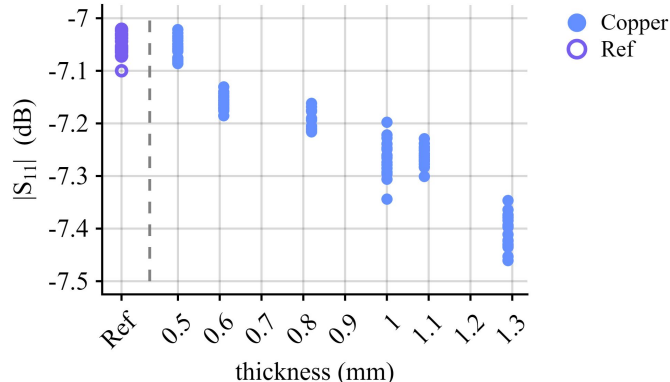


No Relevant Conclusions Could Be Made From Reflection Results

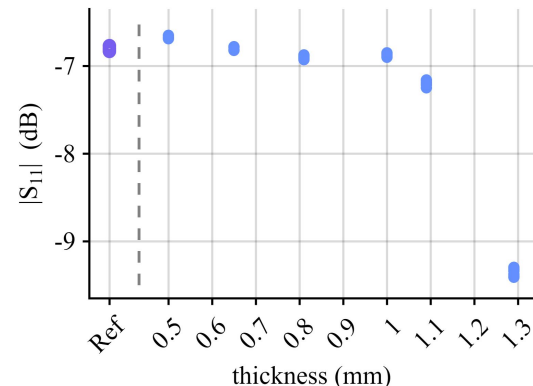
Test 1 - Receiver Reflection



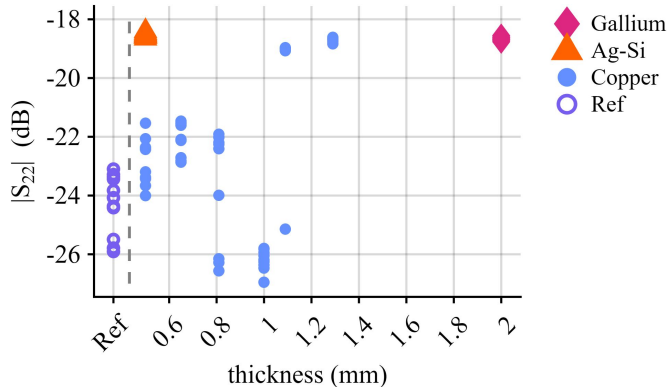
Test 2 - Receiver Reflection



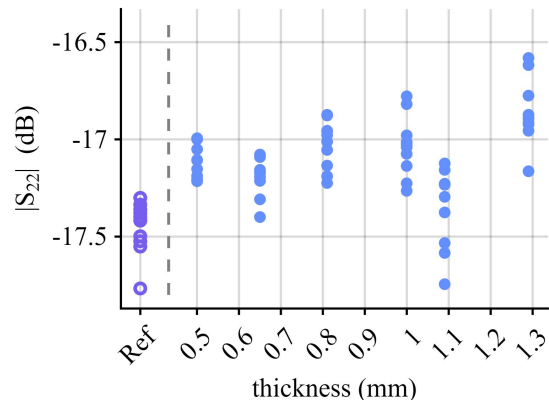
Test 3 - Receiver Reflection



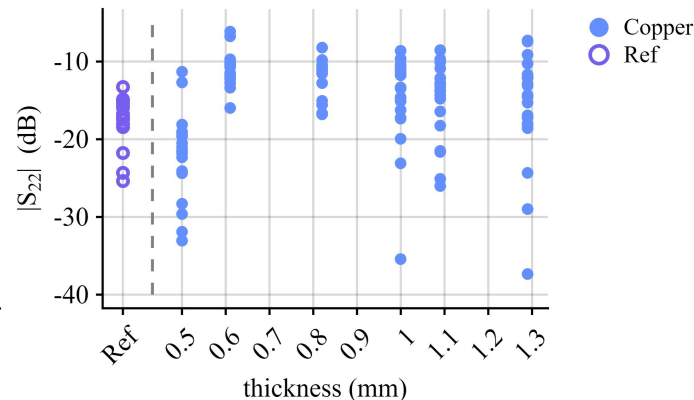
Test 1 - Transmitting Reflection



Test 2 - Transmitting Reflection



Test 3 - Transmitting Reflection

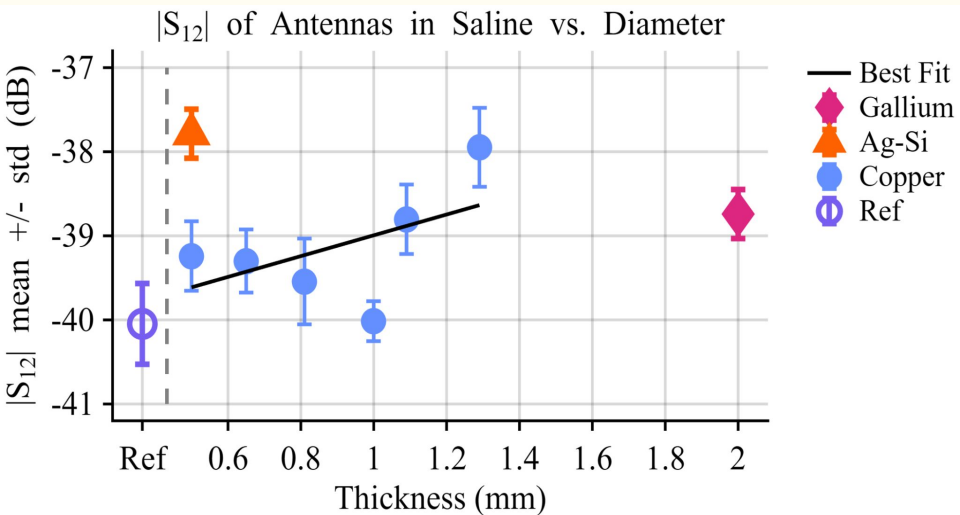


Results from Diameter Tests

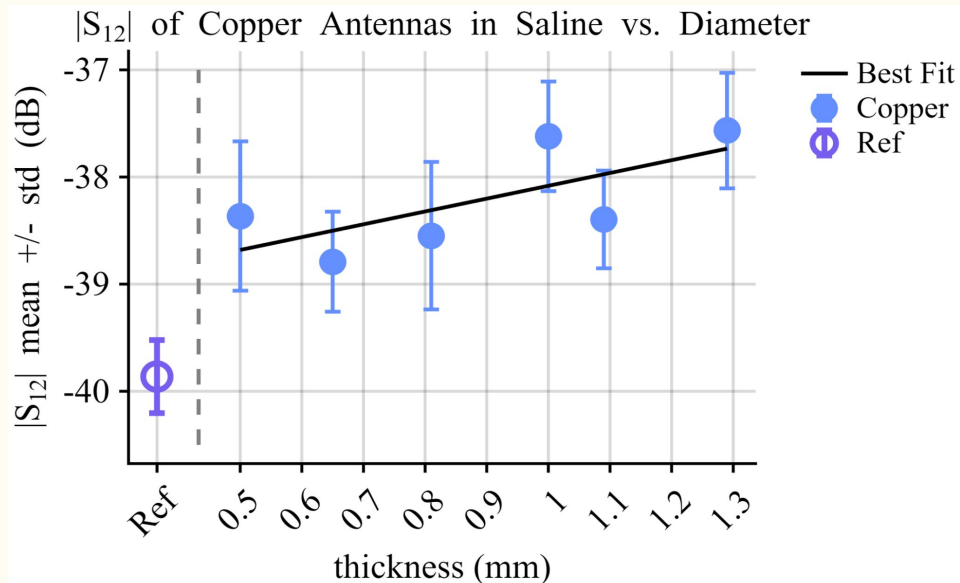
Passive Antenna Diameter Test Results S12 Average and Standard Deviation From Test 1 and 2.

Tests Orientation: 1CM from Receiving Antenna, 1CM Apart at Tip, 45 Degree Angle, 2 inch length.

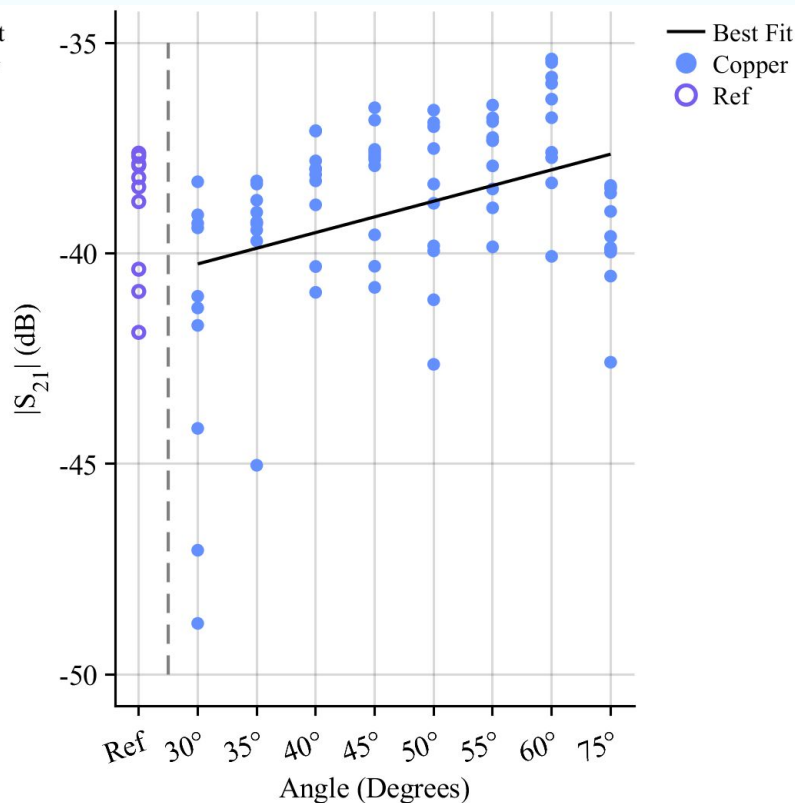
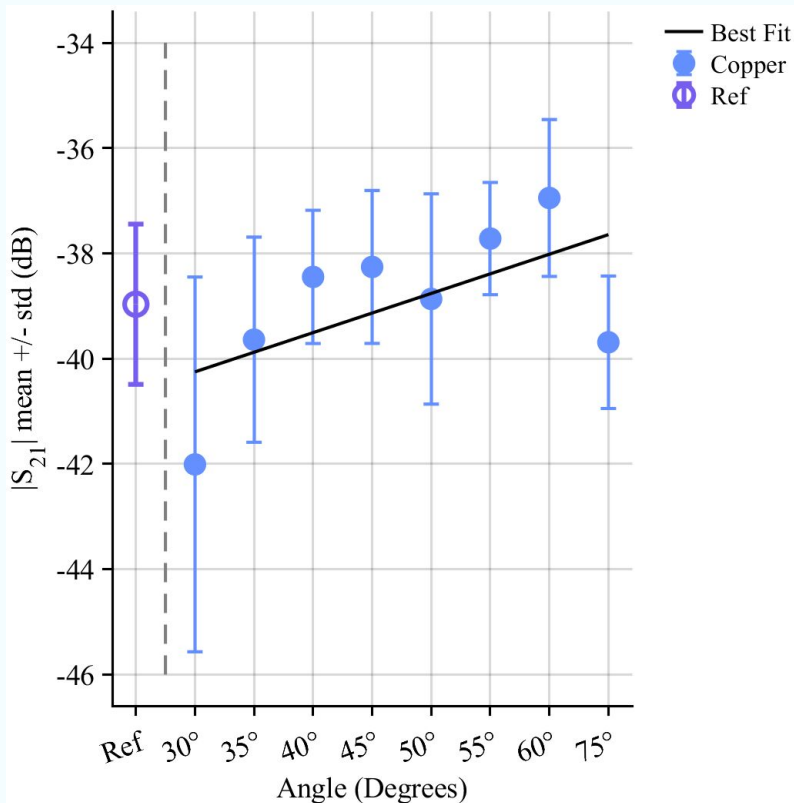
Test 1



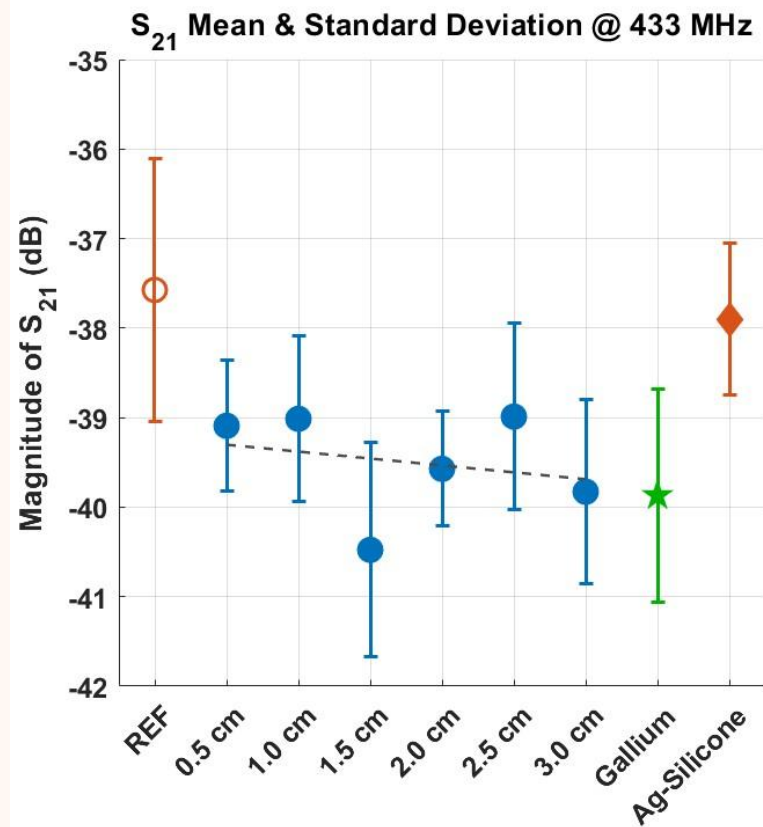
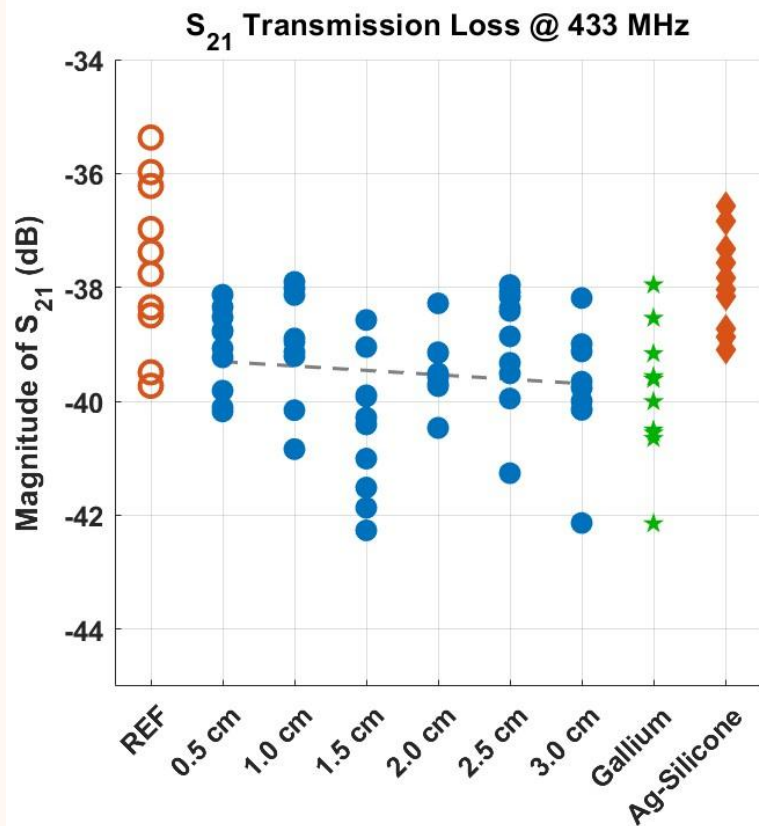
Test 2



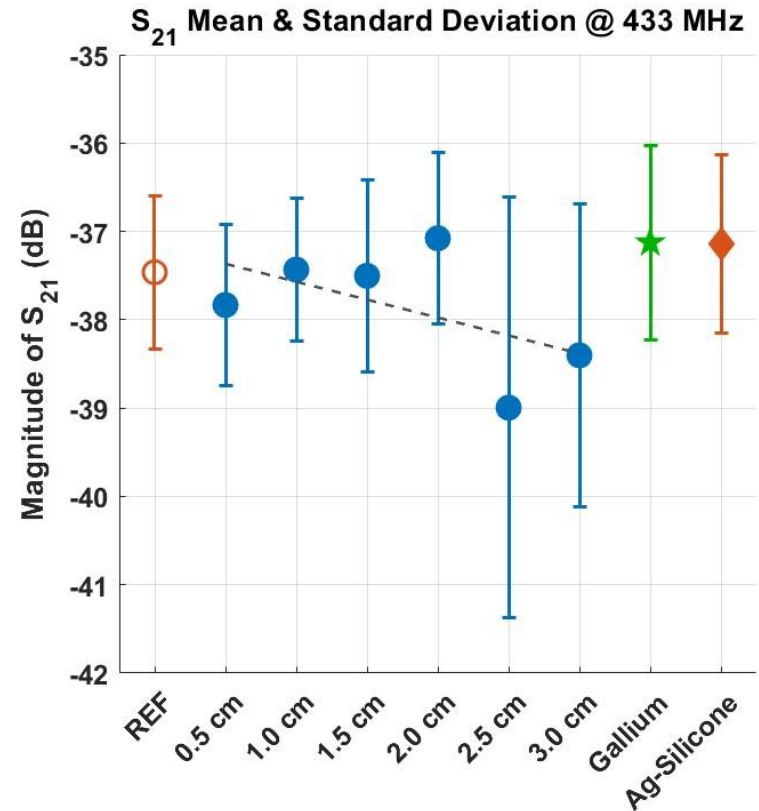
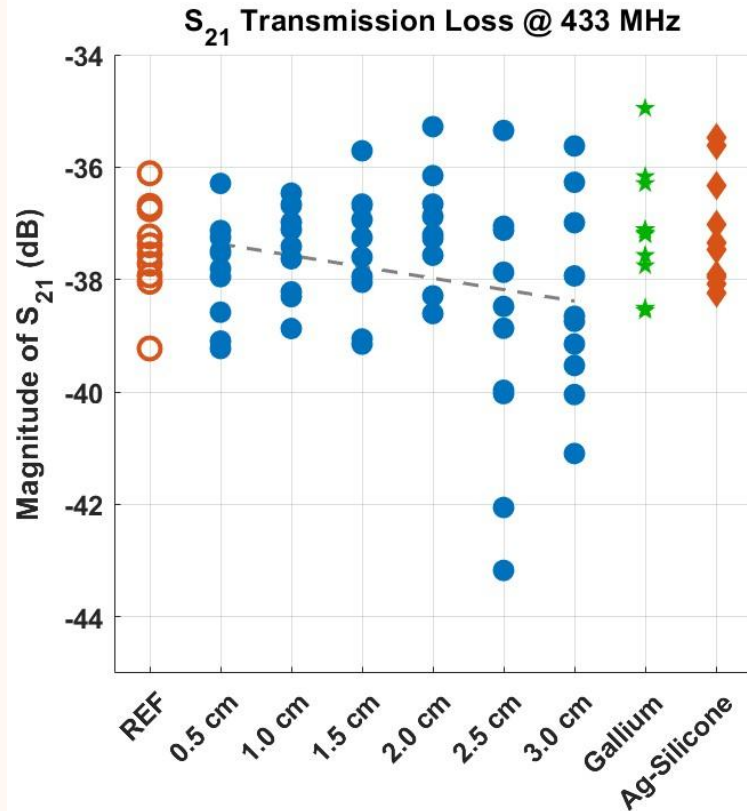
Preliminary Passive Antenna Angle Test Results



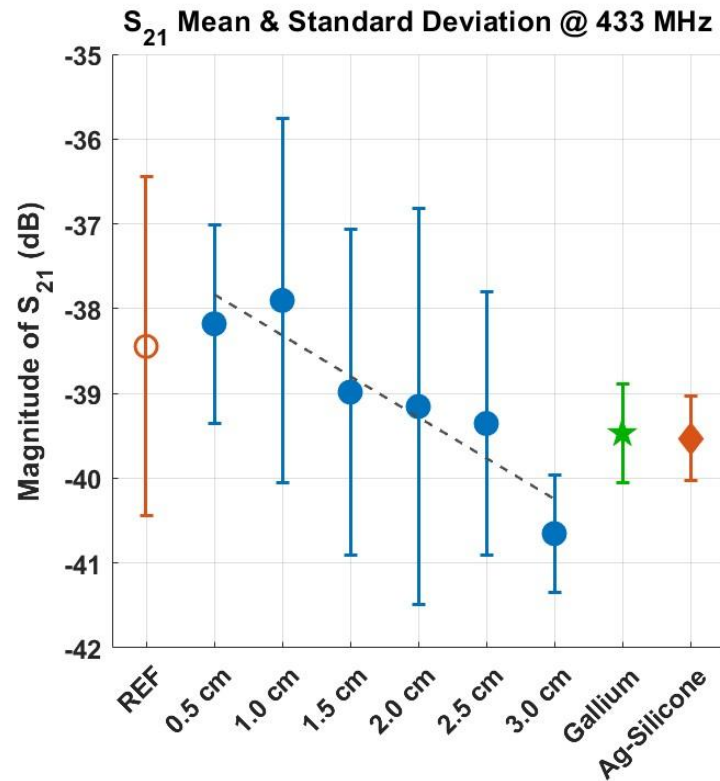
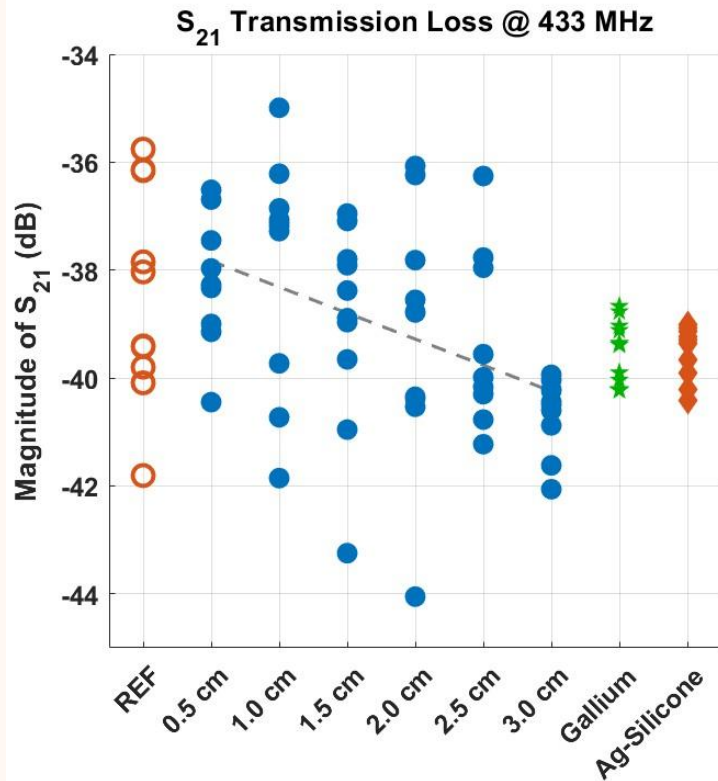
3.18cm Passives @ 45 Deg, Distance to Receiving Antenna Test Plots



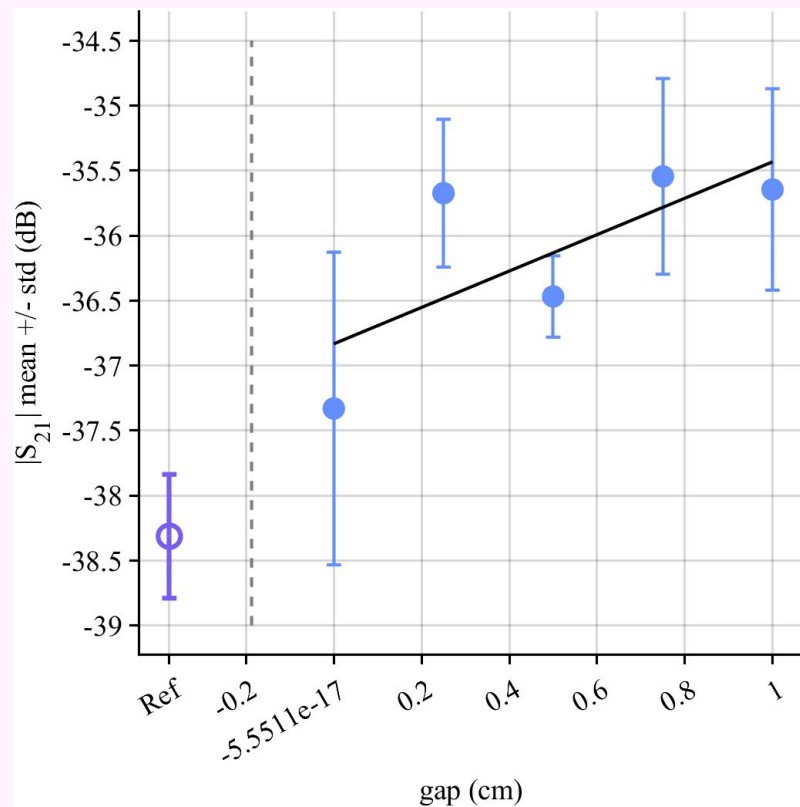
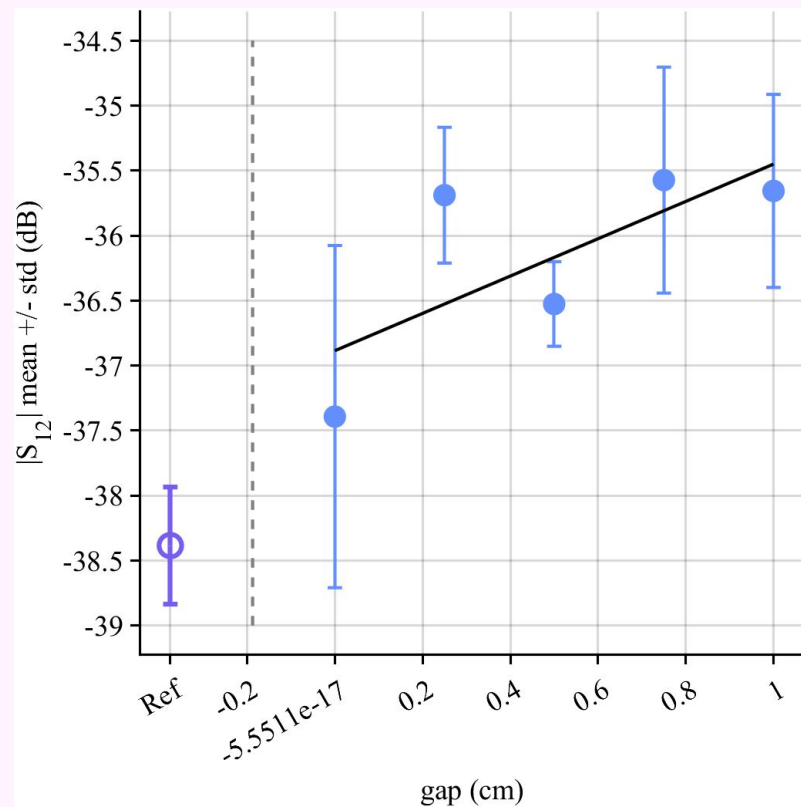
3.8cm Passives @ 45 Deg, Distance to Receiving Antenna Test Plots



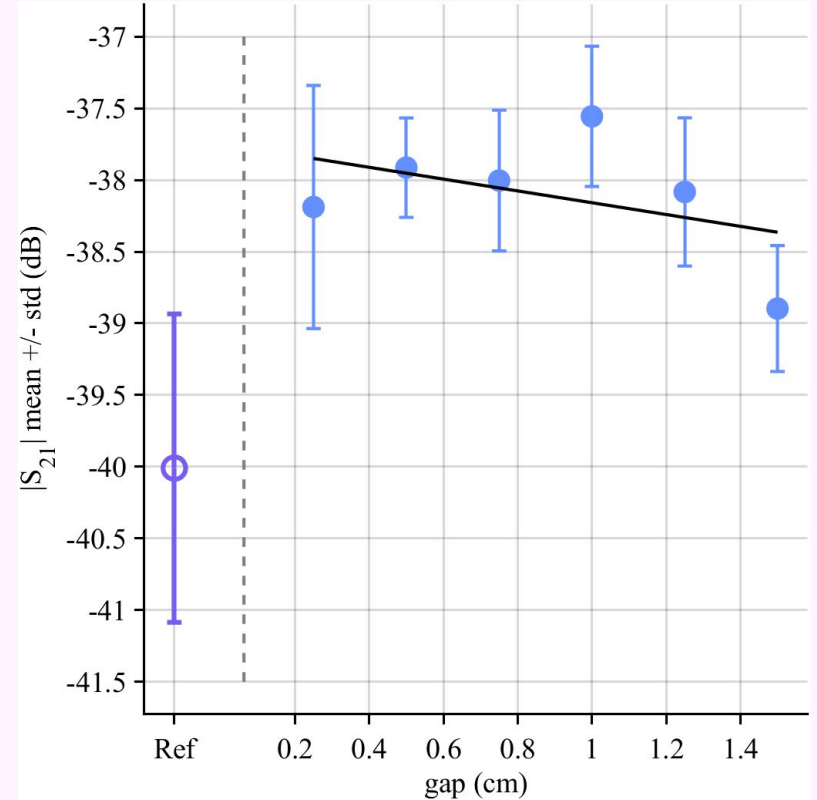
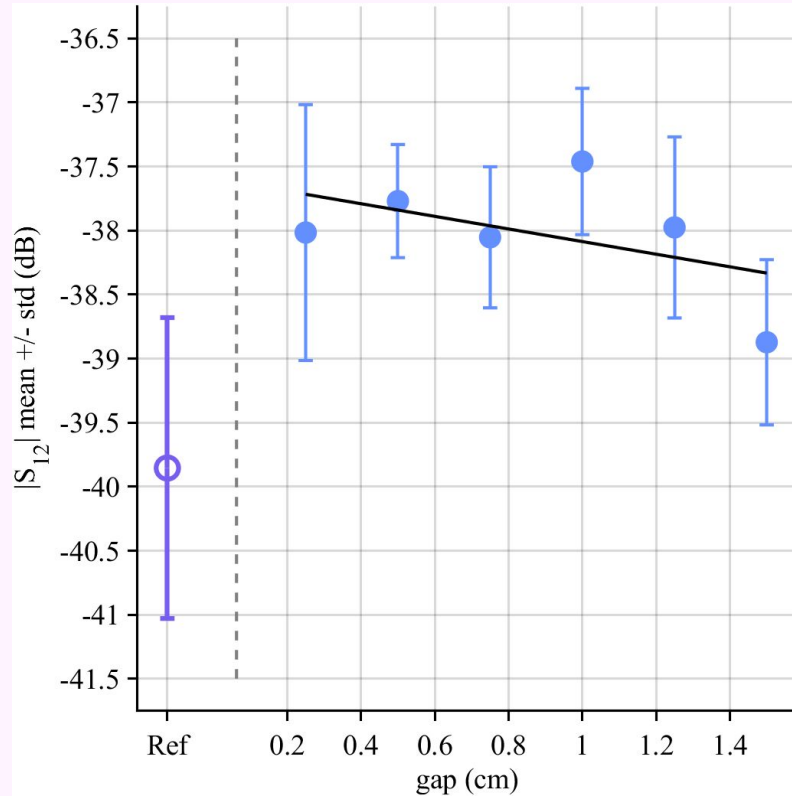
5.1cm Passives @ 50 Deg, Distance to Receiving Antenna Test Plots



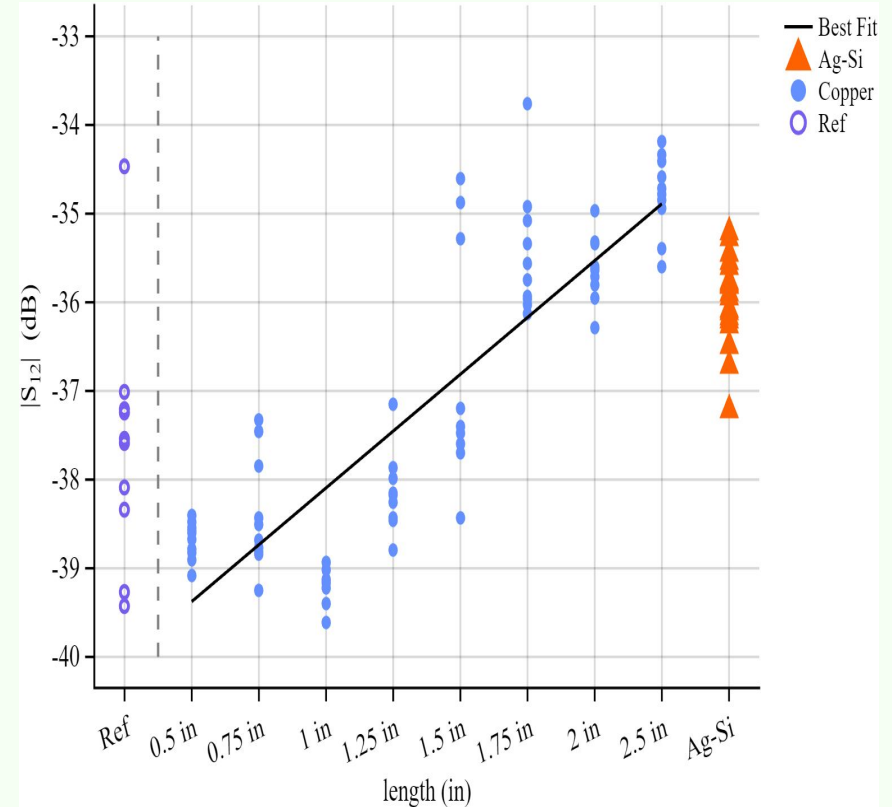
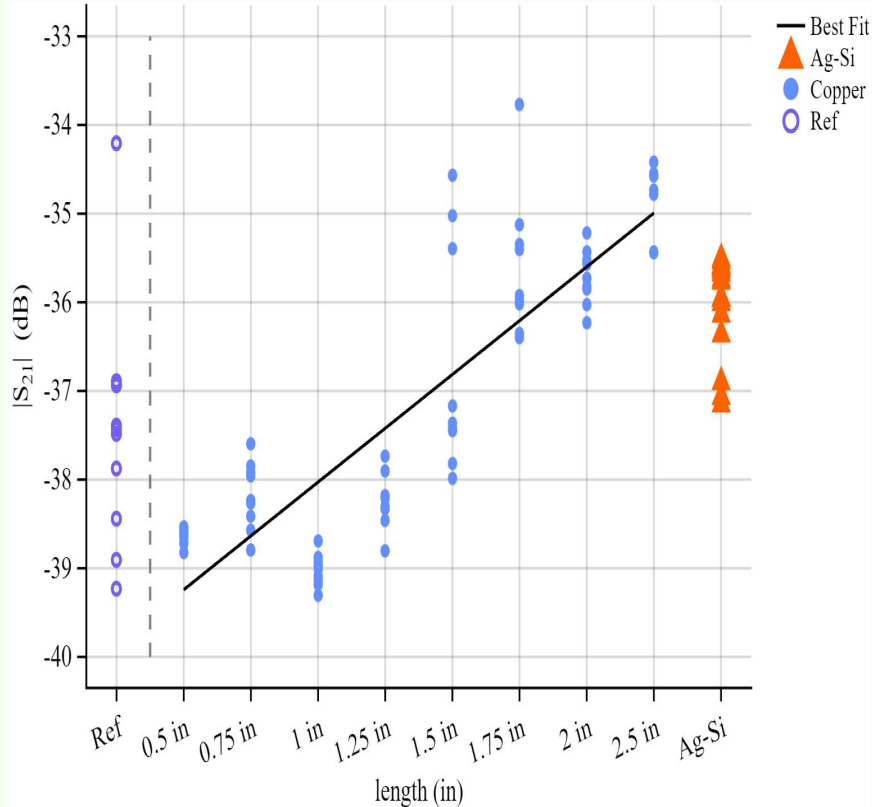
Testing distance between passives, at 45Deg, 0.5cm from receiver, 2 Inches length, 1.29 gauge
Initial tests



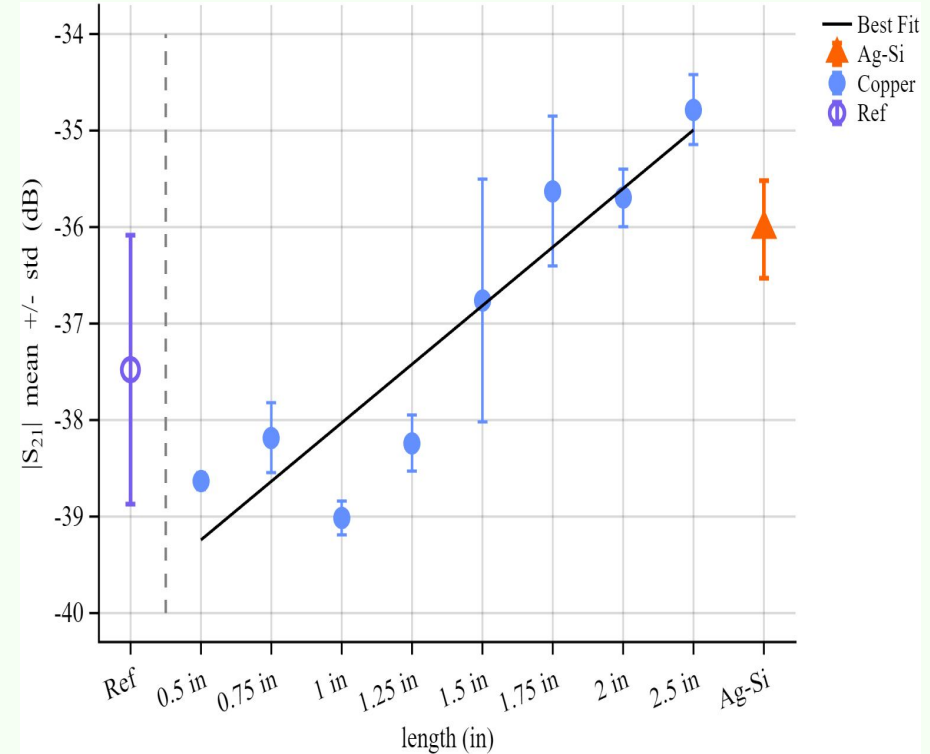
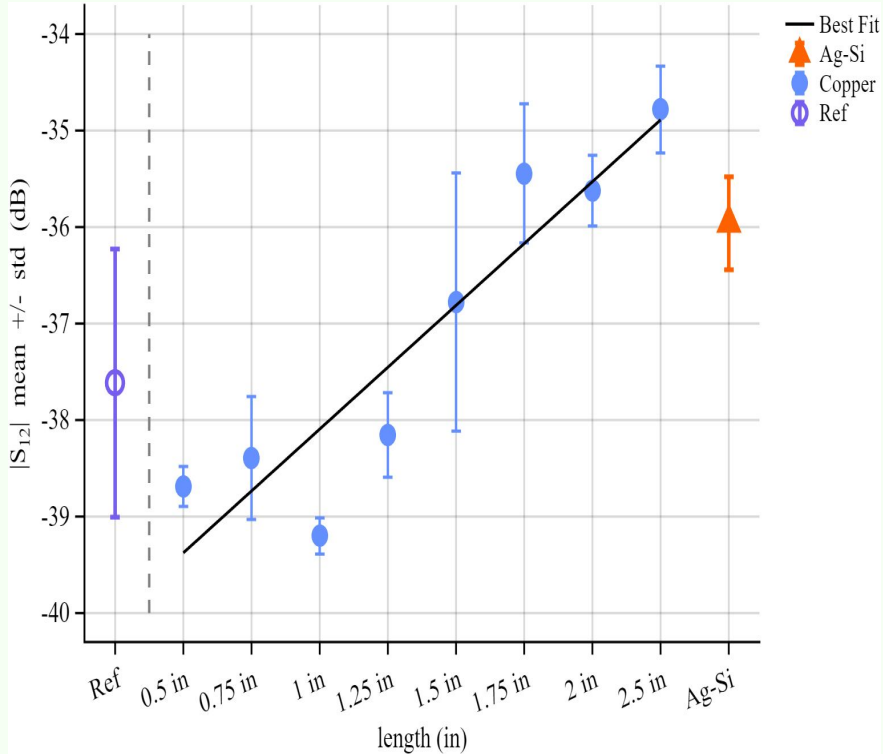
Testing distance between passives, at 45Deg, 0.5cm from receiver, 2 Inches length, 1.29 gauge
Second tests



Length Test Results: S21 & S12 @433MHz, Best Fit



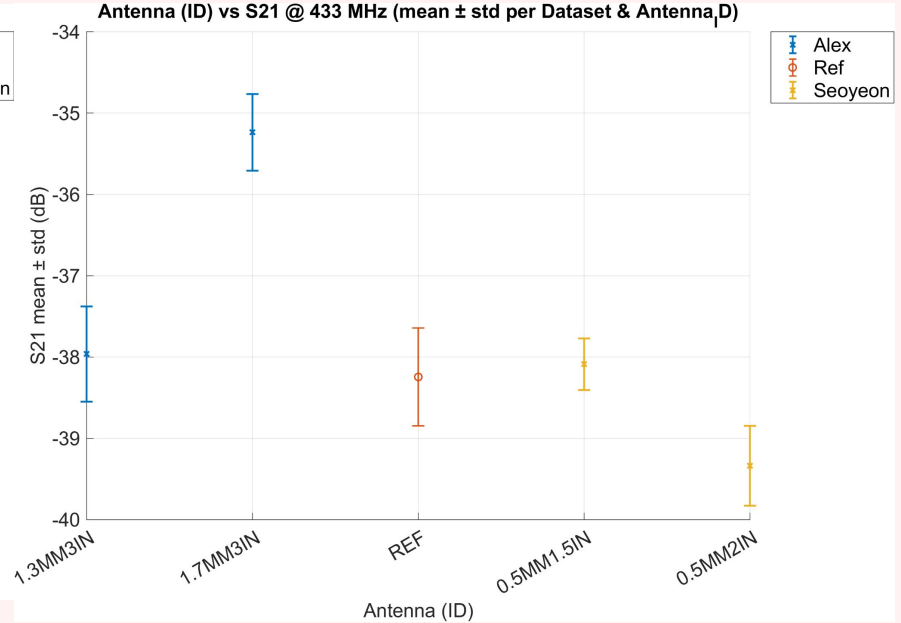
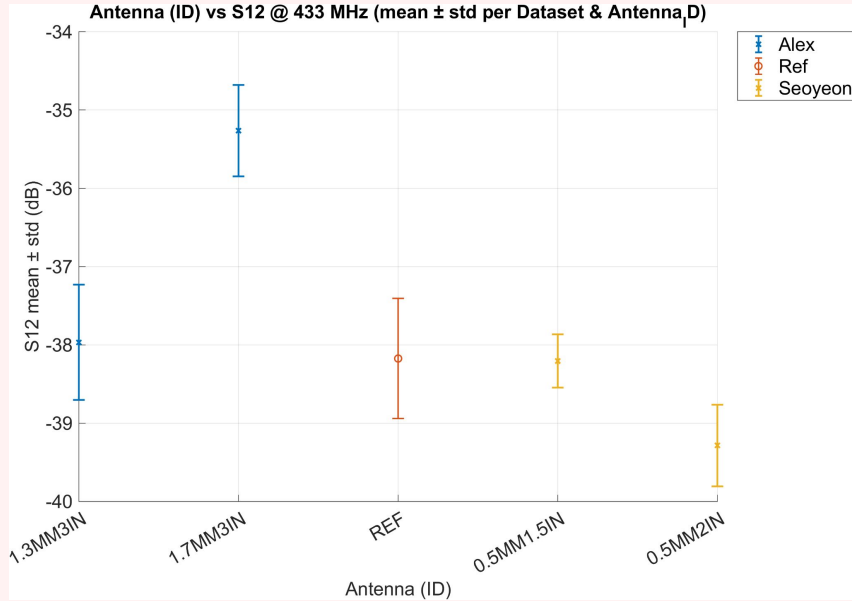
Length Test Results: S12 & S21 @433MHz, Mean + Std



Summary of Test Findings.

- Thicker is better.
- About 55 Degrees is the best orientation.
- Longer is better.
- Closer to the receiving antenna is better.

Quick test of given antennas at 45 degrees



Intermediate Results Testing Copper Passive Antennas with Optimal Parameters from Other Tests

- For this set of tests each parameter was re-measured with the optimal parameters from the previous tests.
- We found these parameters to produce the best transmission results in our previous tests:
 - Longer and thicker passive antennas are better.
 - The distance that the passive antennas are from each other is best at .75cm.
 - Distance away from the receiving antenna was best at .5cm from the top of the receiving antenna.
 - The best angle was 50 degrees.

CAVEAT

- Unfortunately these tests had issues with our VNA calibration making them an unreliable set of test results.
- The Coaxial cables had an intermediate junction with a bad connector compromising these results by increasing the data collection range of each set of individual passive antenna tests.

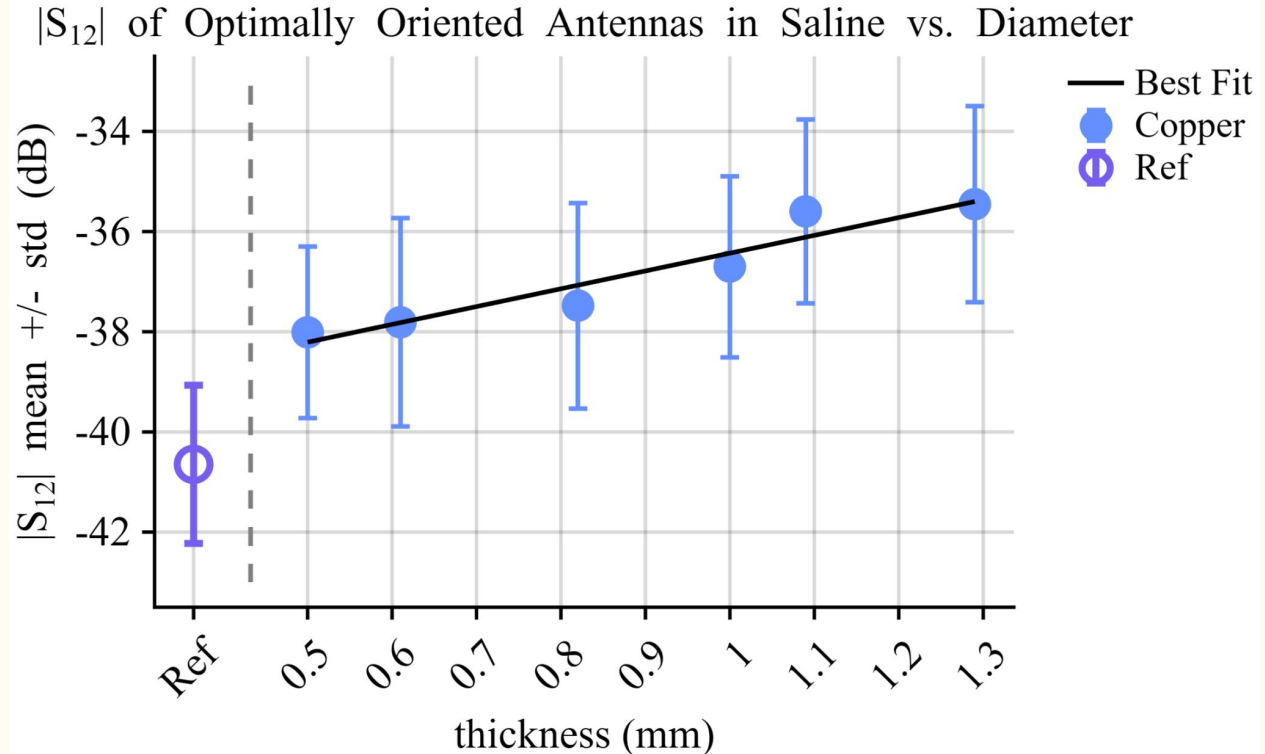
Passive Antenna Thickness (Diameter) Test Results S12

Average and Standard Deviation From Optimal Testing.

Tests Orientation: 0.5 cm from Receiving Antenna, 0.75 cm Apart at Tip, 45 Degree Angle, 2 inch length.

Larger diameter passive antennas result in better transmission between transmitting and receiving antennas.

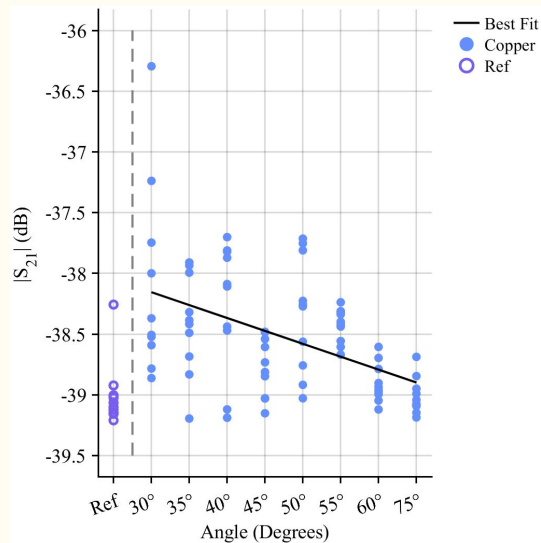
- Diameters larger than 2 mm were not tested.



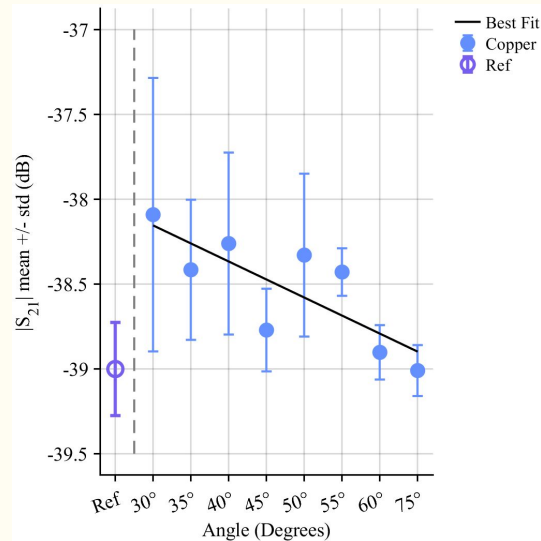
Passive Angle Tests with Optimal Parameters

Passive antennas were 1.29MM, 2 Inches long, .75cm apart, and .5cm from receiving antenna.

All Data Points with Line of Best Fit



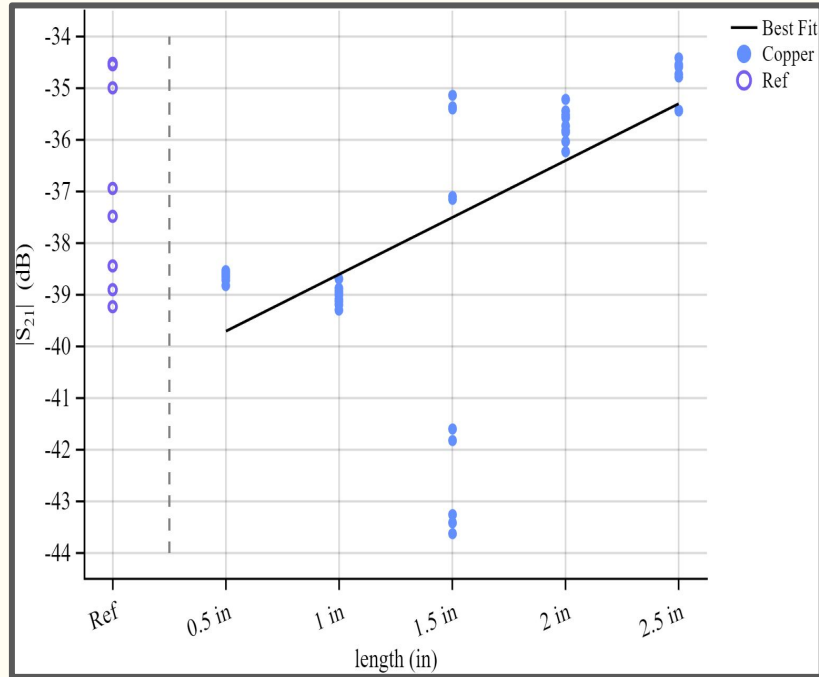
Average and Standard Deviation of Data



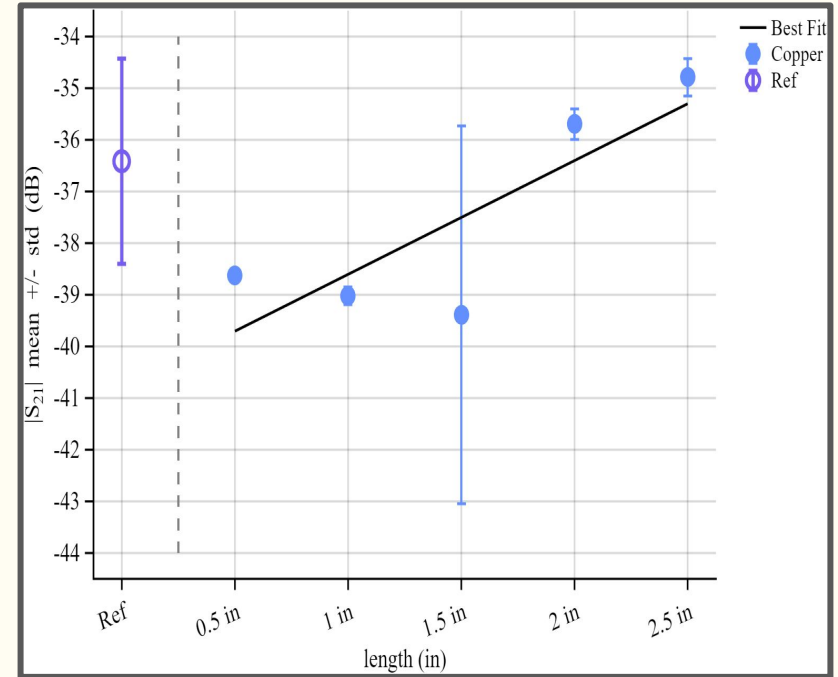
Length With Optimal Parameters

Varying length, @50 Degrees, 1.29mm Thickness, Antennas were 0.75cm Apart & 0.5cm From Receiver

All Data Points with Line of Best Fit



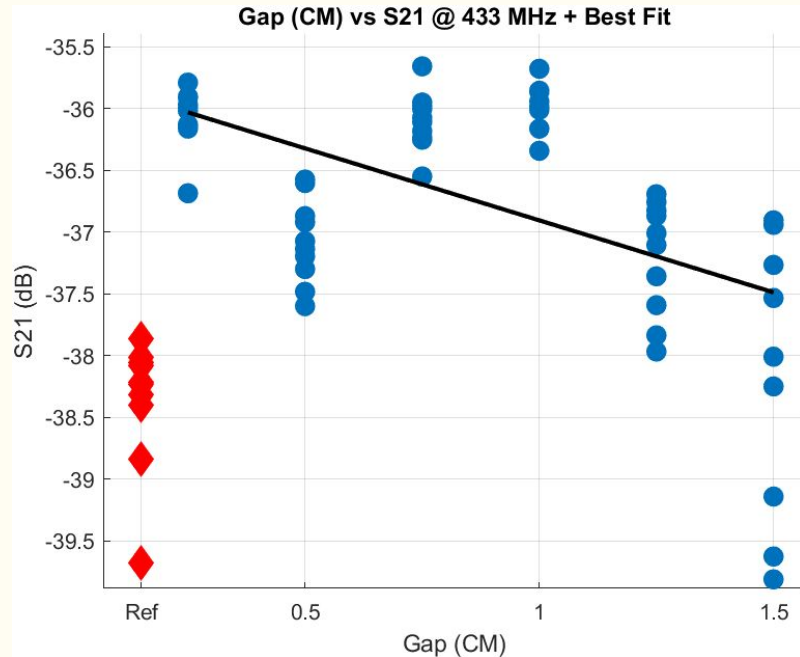
Average and Standard Deviation of All Data



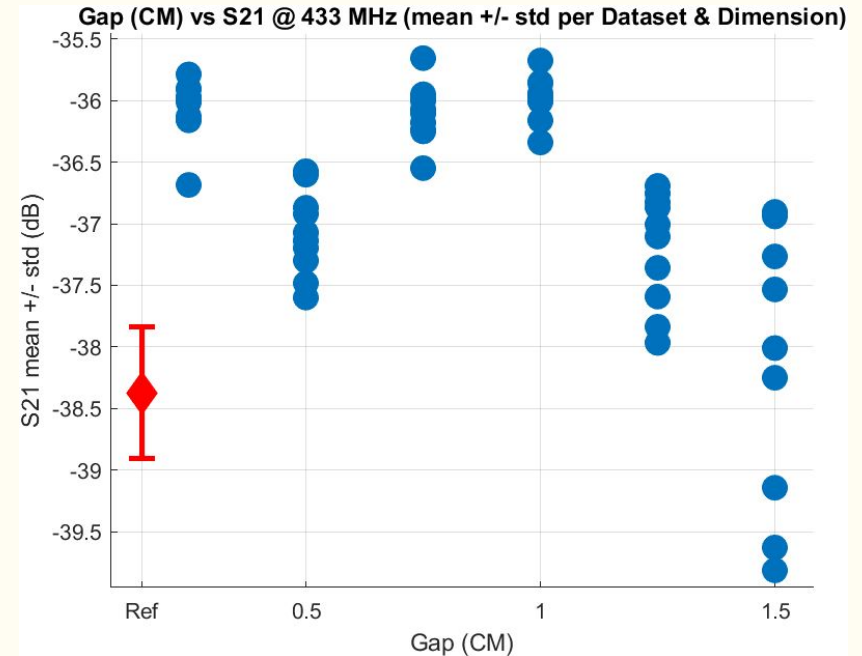
Gap With Optimal Parameters

Tests Orientation: 0.5CM from Receiving Antenna, 50 Degree Angle, 2.5 inch length.

All Data Points



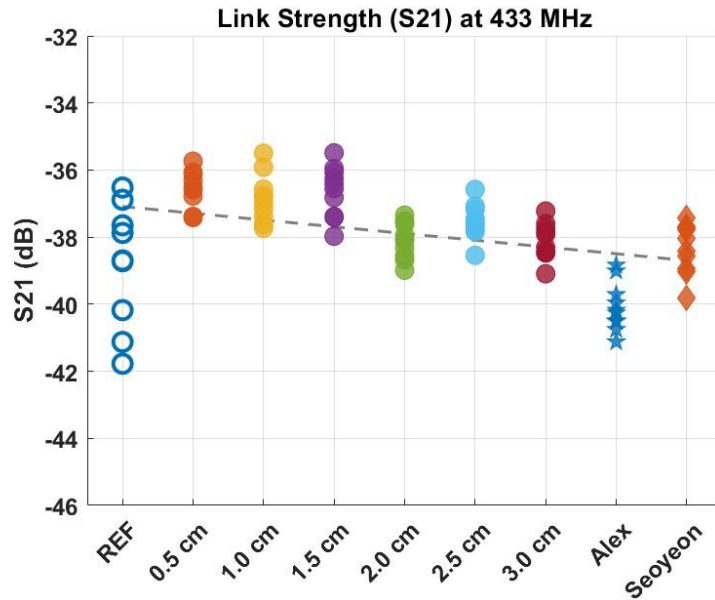
Average and Standard Deviation of All Data



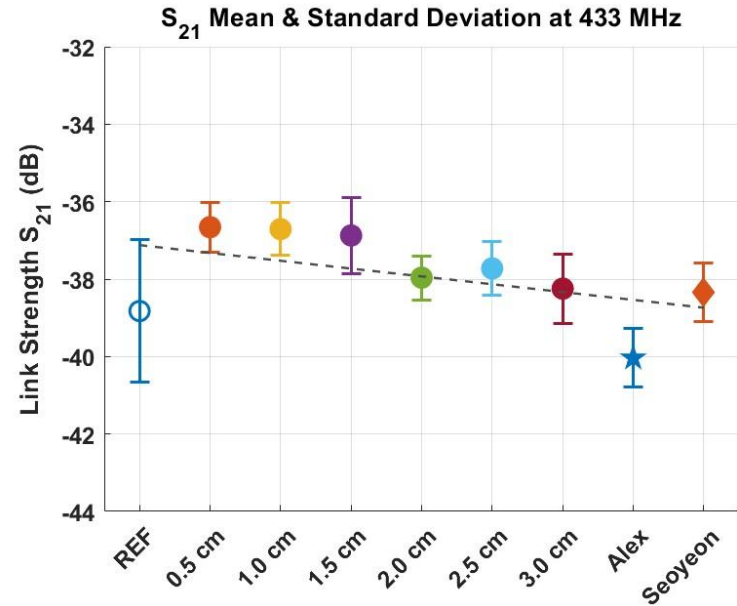
Distance Test With Optimal Parameters

Passive Antennas were 2.0 inches long @ 50 Degrees with 1.3mm thickness

All Data Points with Line of Best Fit



Average and Standard Deviation of Data



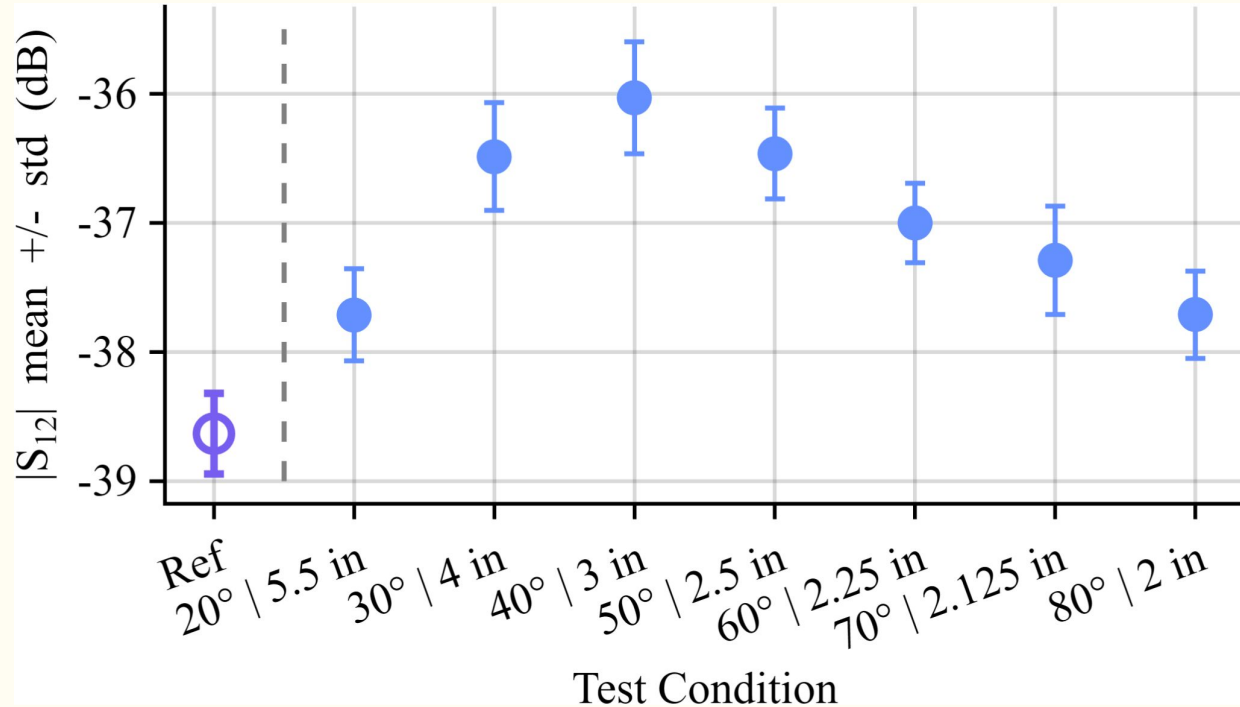
Intermediate Results Testing Copper Passive Antennas with Optimal Parameters from Other Tests

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 - Distance away from the receiving antenna was best at .5cm from the top of the receiving antenna.
 - The best angle was 50 degrees.

Additional Testing of Angle vs. Length of Copper Passive Antennas in Saline

Tests Orientation: 1.29 mm Diameter, .75 cm dist. to receiver, .5 cm gap, longest possible antenna for angle

These results showed that the angle does have an effect on the transmission even as the length gets longer.



Pork Testing

- Experimental Setup
 - All Tests used Nick's wooden Antenna Placement Guide where antennas are spaced 1 cm apart
 - Tests were taken at 50 degrees, 2 inches apart
 - Pork was heated to around 100°F + for 4 hours
 - Silver Flake Antenna and Gallium were pushed into needles, then put into pork using jig. A small wire then pushed the inserted antenna out and into the pork
 - PNIPAM was Injected using a needle
- Caveat
 - Our reference measurements were not taken before inserting passive antennas which most likely altered our results

Image of Pork Testing Set-Up

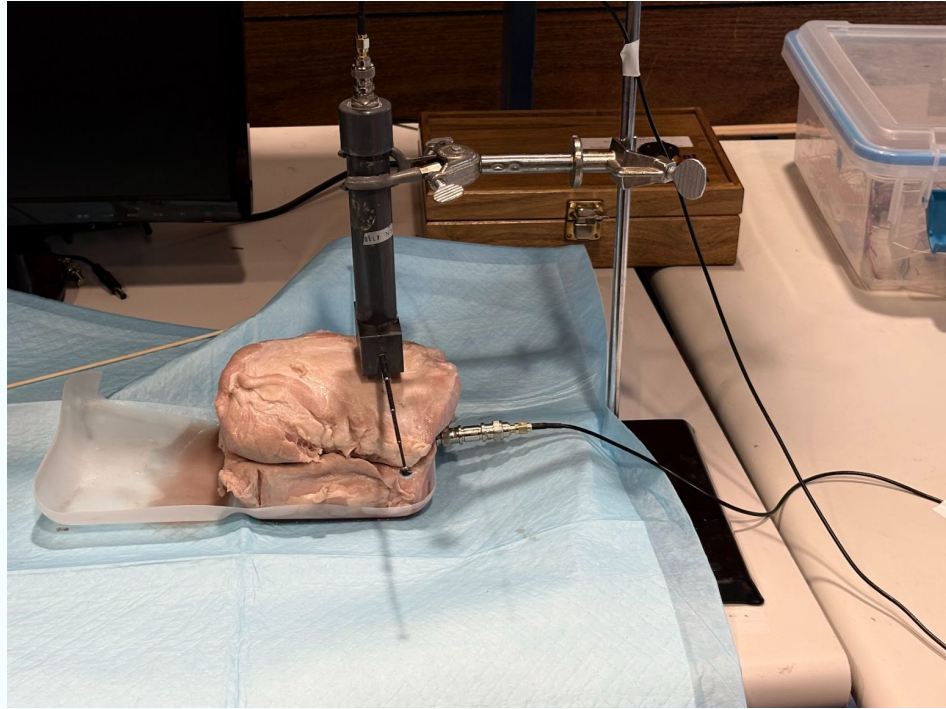
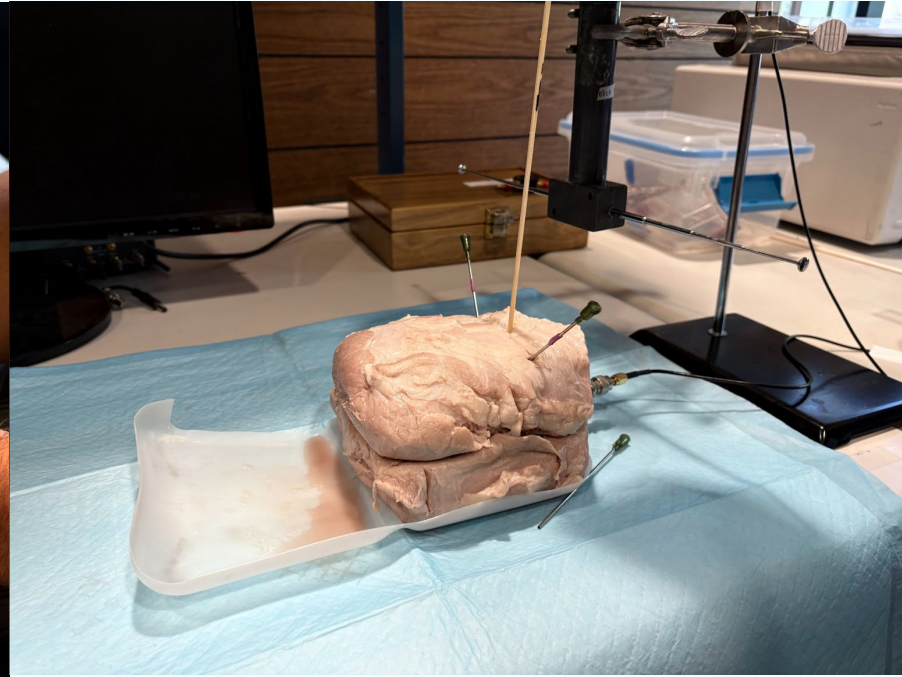
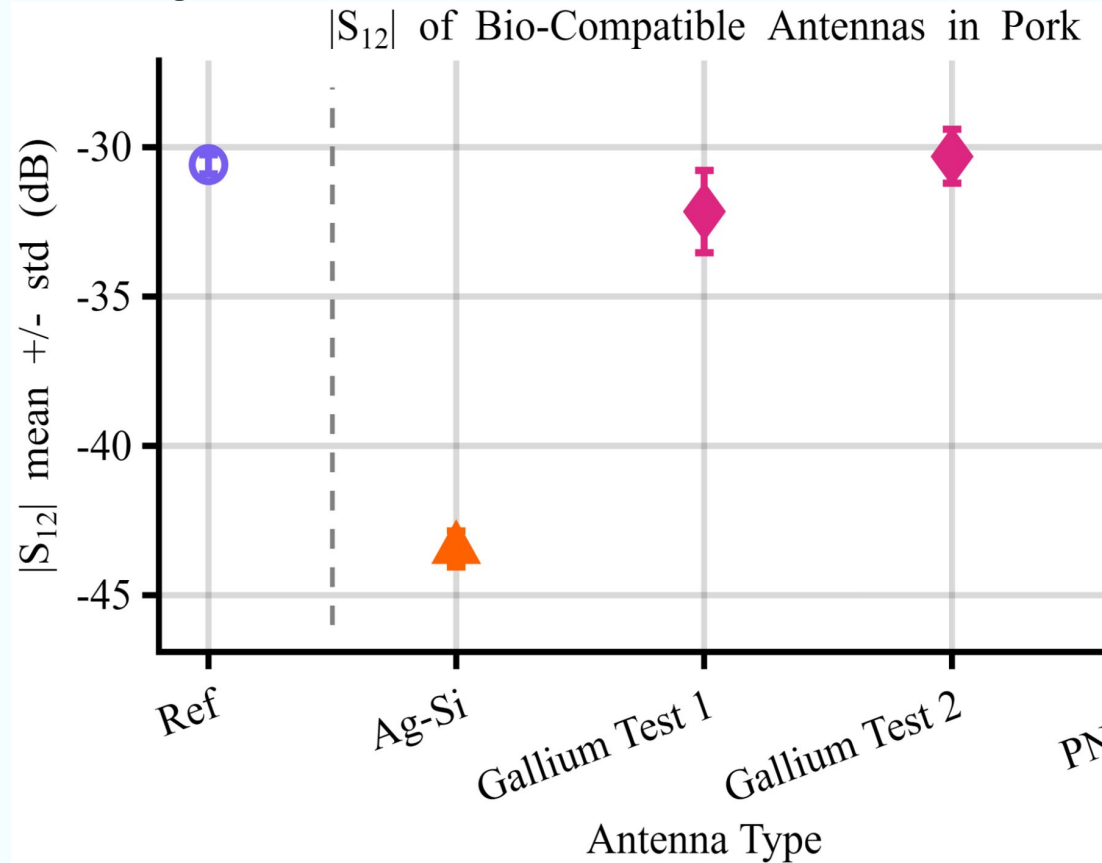


Image of Pork Testing & Antenna Placement Guide In Use



Pork Testing Results

Only the Gallium Tests 2 had relevant reference measurements as we forgot to take reference measurements before the other tests.



Material Performance

PNIPAM

- Produced the best transmission results.
- Formed a consistent, elongated (“booger-like”) structure when solidified
- One of the passive antennas did not fully inject and remained closer to the subsurface level.

Silver Flakes

- Difficult to inject into pork.
- Became compressed/squashed during insertion, likely degrading performance.
- Measurements were inconsistent, especially when PNIPAM was also present (interference/confounding results).

Explanations

- Increased pork thickness → greater signal attenuation
 - Grant's hypothesis:
 - Using less meat may allow the signal to wrap around edges, reducing attenuation
- Measurement observations:
 - Antenna placed at center of pork:
 - Signal propagates through more material
 - Results in ~10 dB decrease in S11
 - Antenna placed near edges:
 - Less material in signal path

Notes

- Mechanical Limitations
 - Using the antenna placement guide to insert passive antennas:
 - Compresses the pork.
 - Causes antenna displacement, reducing placement accuracy.

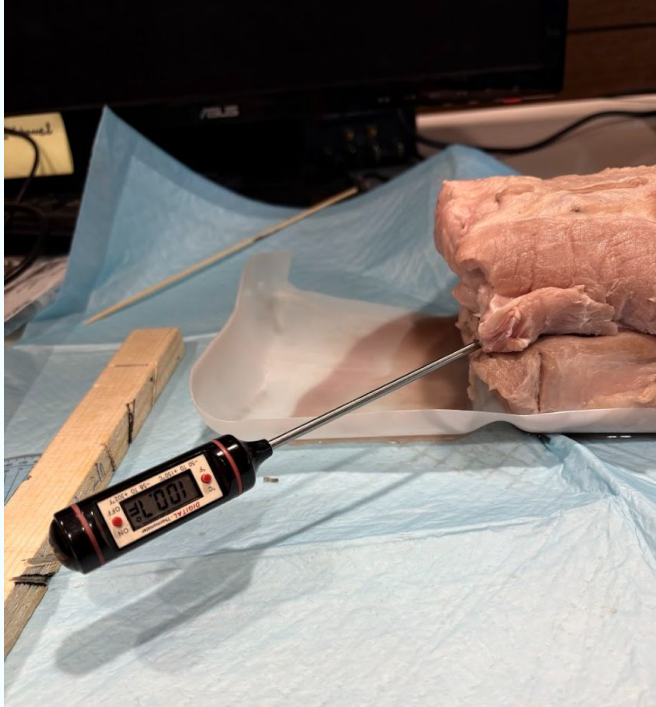
Pork Cross Section & Antennas Compared to Antenna Placement Guide Ideals



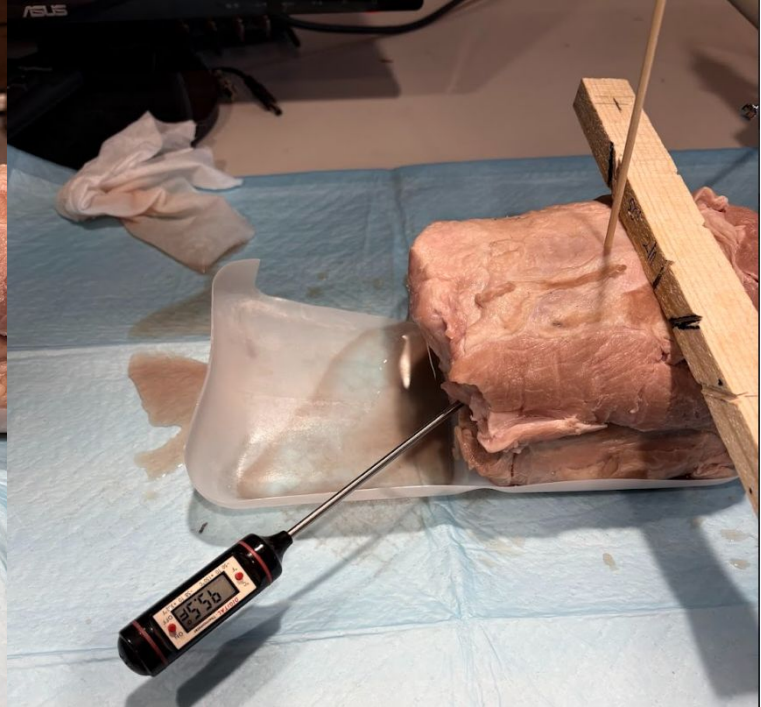
Notes

- Thermal Observations
 - When stacking two warm pork samples:
 - Initial internal temperature: 100.4°F
 - After ~30 minutes: 96.9°F
 - Indicates gradual cooling
- Environmental / Experimental Control
 - Pork should be double-bagged to:
 - Prevent leakage (“pork water”) into the cooler.
 - Avoid moisture contamination affecting measurements.

Temperature of Pork



100.7°F



95.5°F

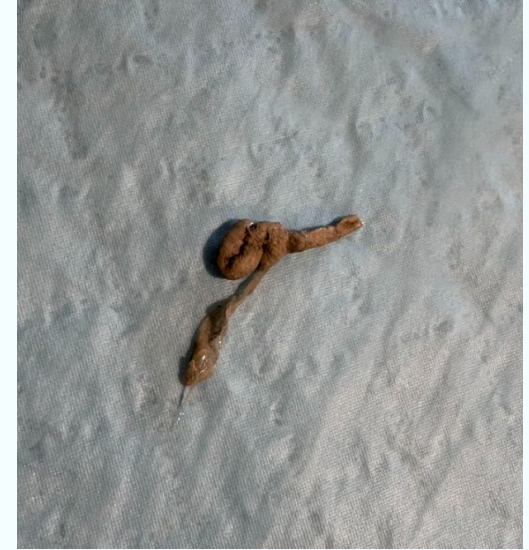
Retrieval of Antennas



Silver Flakes



PNIPAM In Pork



PNIPAM Out
of Pork

Improvements for the Future

- Pork injection procedure needs to improve to achieve optimal placement.
- Reference Testing Needs to occur before each test because the meat can be different for each test.